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Unit 2 Project Submission

Exercise 1: Summarize appropriate use of cryptography in remote access

FIC Bank's internal network relies on secure remote access to support communication among its employees. In remote access scenarios, FIC Bank uses cryptography to protect sensitive data as it travels between remote users and the internal network. This process allows the financial organization to support four main security goals: confidentiality, integrity, authentication, and non-repudiation (SecurityScorecard, 2025). It is important for the security team to implement encryption to ensure that data cannot be read or altered by unauthorized parties if traffic is intercepted during transmission. To maintain the appropriate use of cryptography, sensitive data is classified into five categories, allowing encryption controls to be applied based on the level of risk and data sensitivity. While open or public internal information, such as general new employee announcements, may not require encryption, sensitive information such as commercial agreements with third parties and potential merger or acquisition data requires strong encryption to maintain confidentiality and reduce organizational risk (Longbottom, 2018).

Based on the remote access scenario, different encryption methods are used depending on the purpose and sensitivity of the data. These methods include symmetric encryption, asymmetric encryption, hash functions, and digital signatures. Symmetric encryption uses a single shared key to encrypt and decrypt data and is commonly used for stored data and large volumes of information. During active VPN sessions, FIC Bank uses symmetric encryption to protect internal system communications and shared database traffic because it provides strong security while maintaining performance. Asymmetric encryption uses a public and private key pair and is primarily used during the authentication and key exchange process. At FIC Bank, asymmetric encryption is used to establish VPN connections and verify digital certificates through Public Key Infrastructure (PKI). This method ensures secure identity verification, while symmetric encryption maintains efficiency once the secure connection is established. Hash functions and digital signatures also play an important role in FIC Bank's security network. Hashing algorithms convert data into a fixed-length value, which helps protect passwords, documents, and messages by ensuring data integrity (SecurityScorecard, 2025). Digital signatures combine asymmetric encryption with hashing to verify the origin and integrity of data, ensuring it has not been altered during transmission (SecurityScorecard, 2025). FIC Bank relies on these cryptographic methods for financial transactions and secure document signing by employees and customers. Together, these core cryptographic techniques support secure remote access by protecting identity, data integrity, and performance across the organization's network.

Overall, FIC Bank should base its encryption preferences on data classification. Highly sensitive data, such as customer financial records accessed remotely by FIC Bank employees, requires strong encryption standards to meet regulatory and security requirements. Less sensitive data may still be encrypted but may not require the same level of cryptographic complexity. If data classifications are not properly applied, it can cause serious technical errors. It can result in false positives and false negatives, such as information that should be classified as "for your eyes only" being incorrectly classified as open and not triggering the appropriate security controls. To

reduce these risks, it is better to classify new documents at creation rather than relying on existing classifications, ensuring encryption is applied correctly from the start (Longbottom, 2018).

While stronger and more complex encryption improves security, it also introduces drawbacks. Advanced cryptographic methods can increase processing time, add network latency, and place additional strain on system resources. In a global organization like FIC Bank, overly complex encryption could impact performance for remote users, especially when accessing systems across different regions. For this reason, organizations must balance security and usability by selecting encryption methods that provide adequate protection while maintaining reliable network performance.

By applying cryptography based on remote access context, data sensitivity, and operational needs, FIC Bank can protect sensitive information while ensuring secure and efficient remote connectivity for its workforce.

Exercise 2: Use network administration tools to implement secure remote access controls

Part 1 A: Creating a User Account to Test Access Restrictions

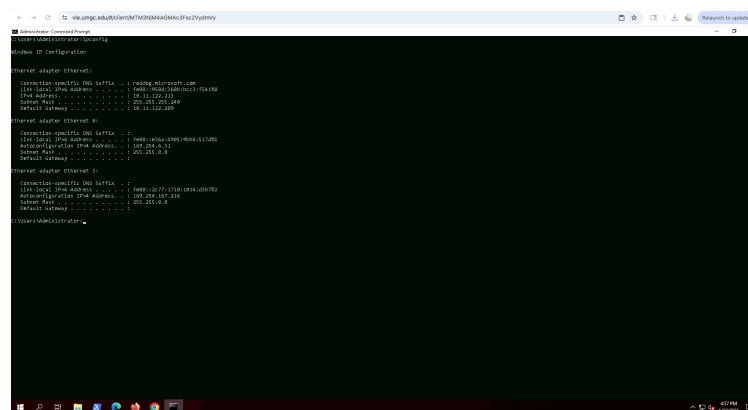
Creating a User Account and Testing Access Restrictions Results:

In this part of the exercise, a new Windows user account named Student1 was created to test remote access permissions. The Student1 account was successfully added through the Computer Management console, and the password was configured according to the lab requirements. After the account was created, Student1 was added to the Remote Desktop Users group to allow remote login access.

Using Remote Desktop Connection, a successful login was performed with the Student1 credentials. Once logged in, a different desktop environment was displayed, and the whoami command confirmed that the session was running under the Student1 account. This verified that the user had proper remote access permissions.

To test access restrictions, Student1 was then removed from the Remote Desktop Users group. After removal, another Remote Desktop login attempt was made using the Student1 account, which resulted in an access denied error. This confirmed that removing the user from the group immediately revoked remote access. Overall, the results demonstrate how group membership directly controls Remote Desktop access on a Windows system. Screenshots Below:

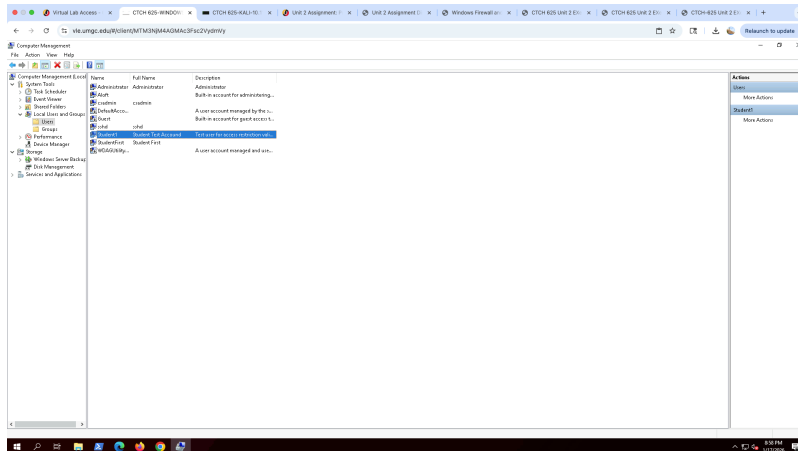
Screenshot 1: Window IP address



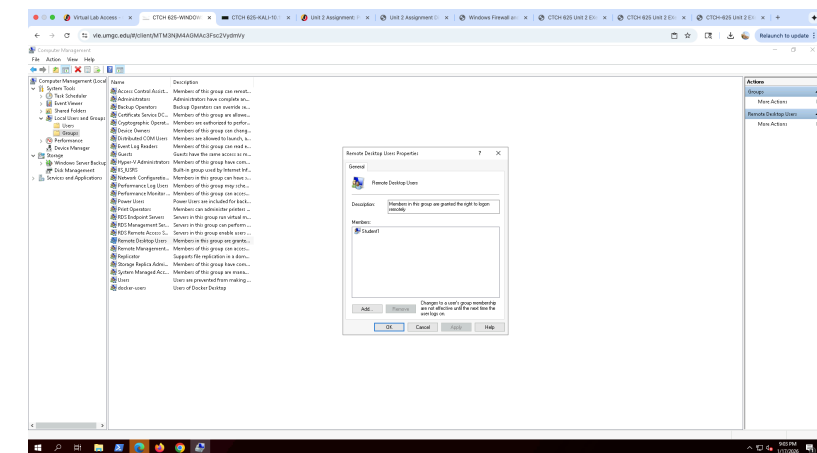
Screenshot 2: Opened Computer Management Admin



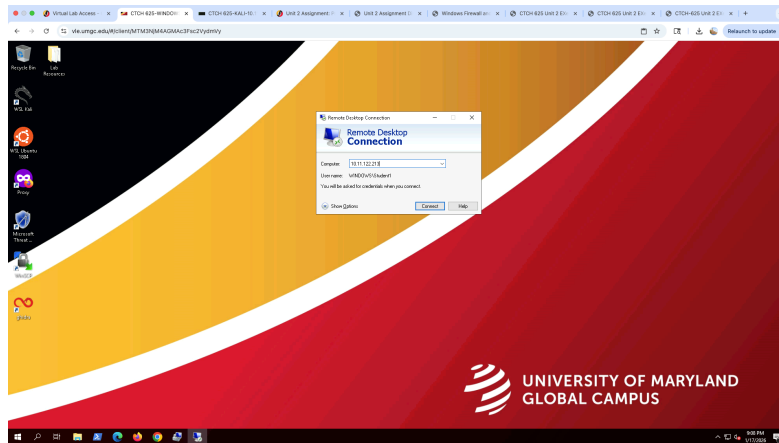
Screenshot 3: Computer Management User Student1 Added



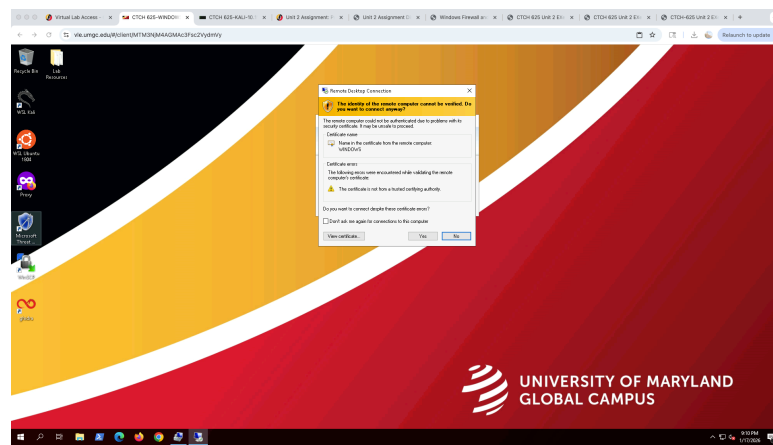
Screenshot 4: Computer Management Group Student1 Added



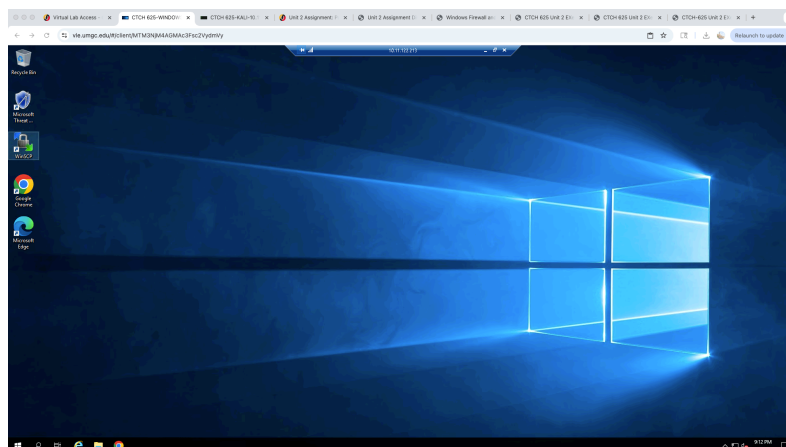
Screenshot 5: Remote Desktop Access



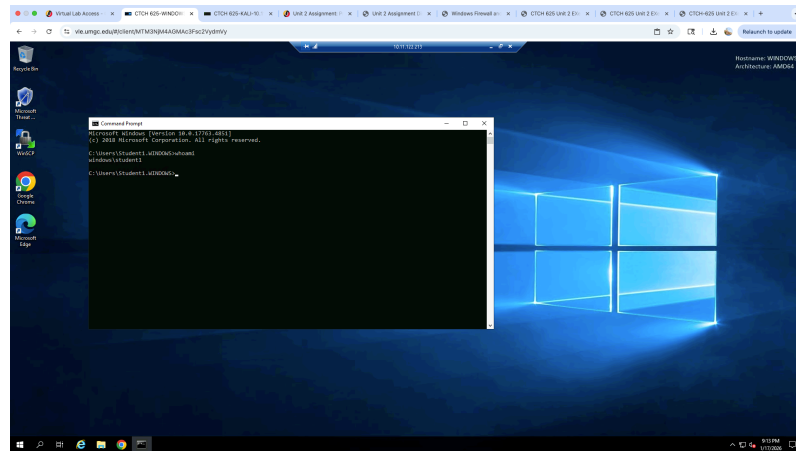
Screenshot 6: Remote Desktop Access Security Prompt



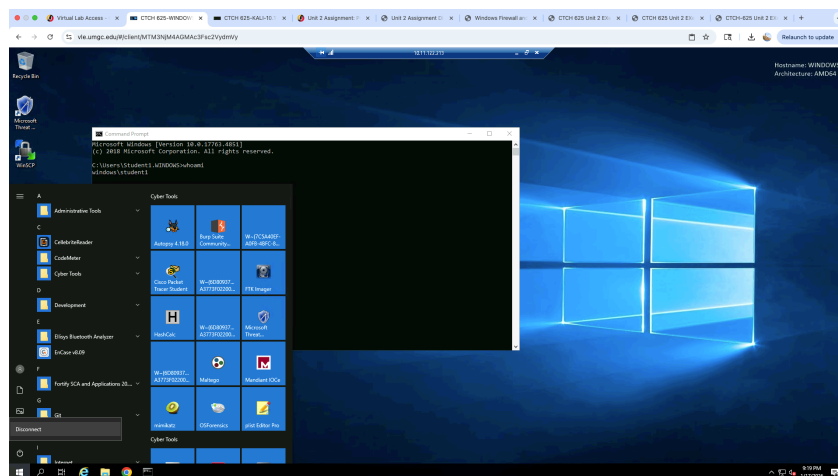
Screenshot 7: Remote Desktop Access Access “Student1” Windows



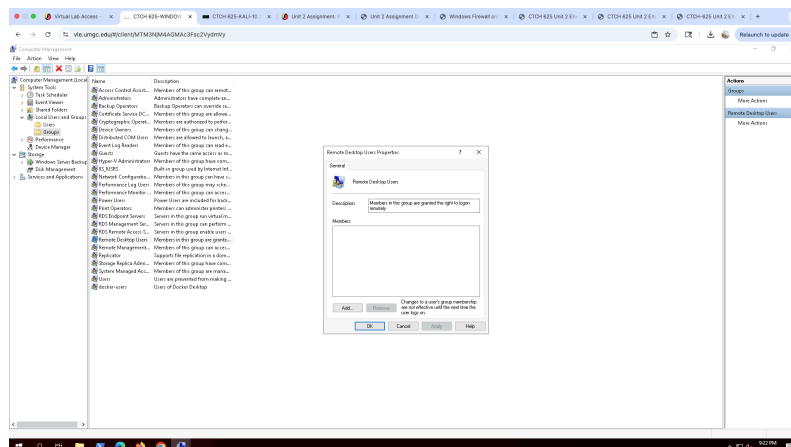
Screenshot 8: Remote Desktop Access Access “whoami” command prompt



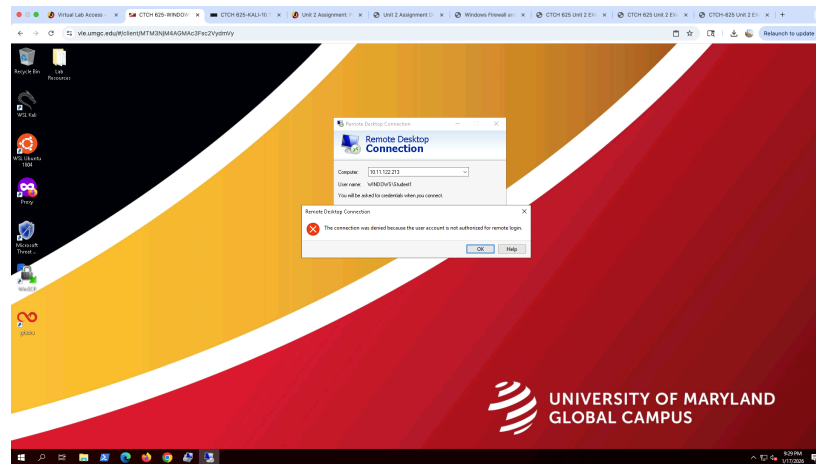
Screenshot 9: Remote Desktop Access Access “discount”



Screenshot 10: Computer Management “student1” Removed



Screenshot 11: Remote Desktop Access Access “student1” Removed



Part 1 B: Managing SSH Access with Windows Firewall and Kali

Part 1: Enable SSH from Kali to Windows

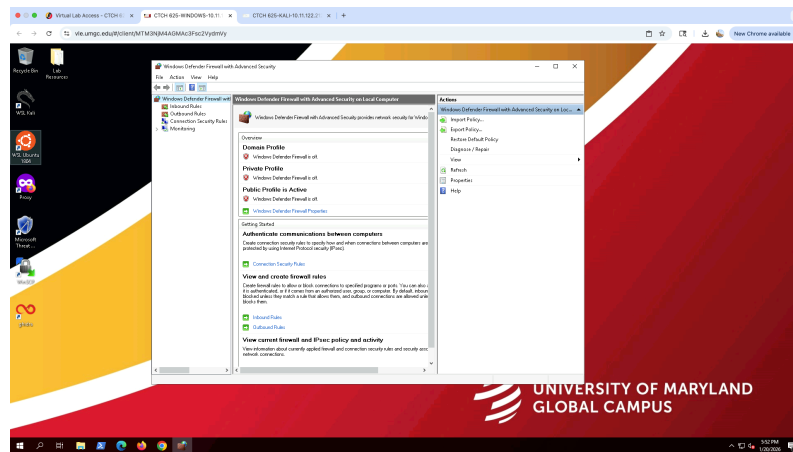
Managing SSH Access from Kali to Windows Results:

In this part of the exercise, SSH access from the Kali Linux system to the Windows virtual machine was successfully restricted using Windows Defender Firewall. First, Windows Defender Firewall with Advanced Security was opened on the Windows VM, and the Inbound Rules section was accessed. The existing OpenSSH SSH Server (Port 22) rule was located and confirmed to be enabled, allowing SSH connections to the system.

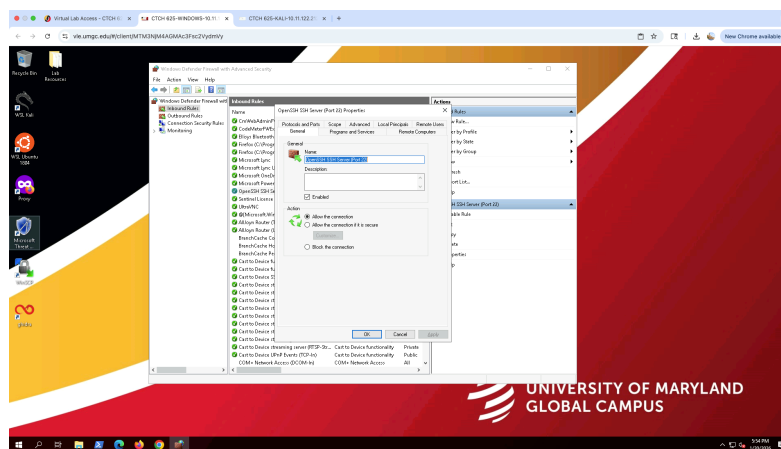
Next, the properties of the OpenSSH rule were modified to limit which systems could connect. Under the Scope tab, the Remote IP addresses setting was changed from allowing any IP address to allowing only a specific address. The Kali Linux IP address 10.11.122.212, identified from the Kali terminal, was added to the allowed list. This ensured that only the Kali system could initiate an SSH connection to the Windows machine.

After applying the changes, the firewall rule was successfully updated and saved. These results confirm that SSH access to the Windows system is now restricted to a trusted source rather than being open to all devices. This demonstrates how Windows Firewall can be used to enforce access control by limiting remote connections based on IP address, strengthening overall system security. Screenshots are below:

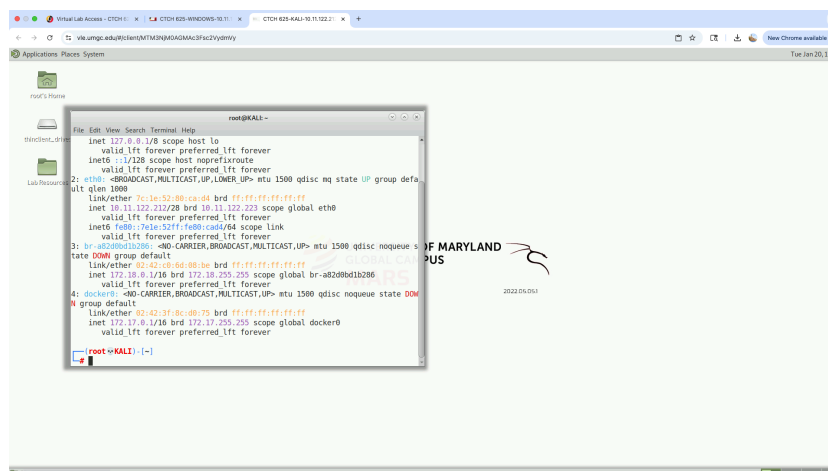
Screenshot 1: Opening Windows Defender Firewall with Advanced Security and accessing inbound rules



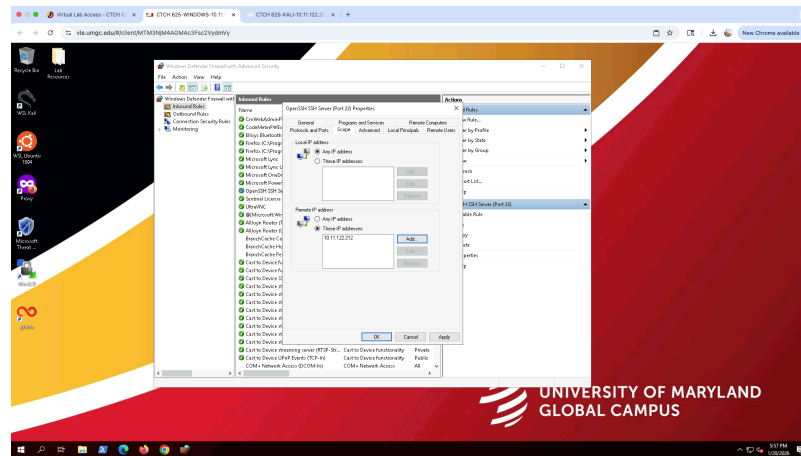
Screenshot 2: Viewing Open SSH Server (Port 22) inbound rule properties



Screenshot 3: Identifying the Kali Linux IP address (10.11.122.212) from Kali Terminal



Screenshot 4: Restricting SSH access by adding the Kali IP address under the Scope tab

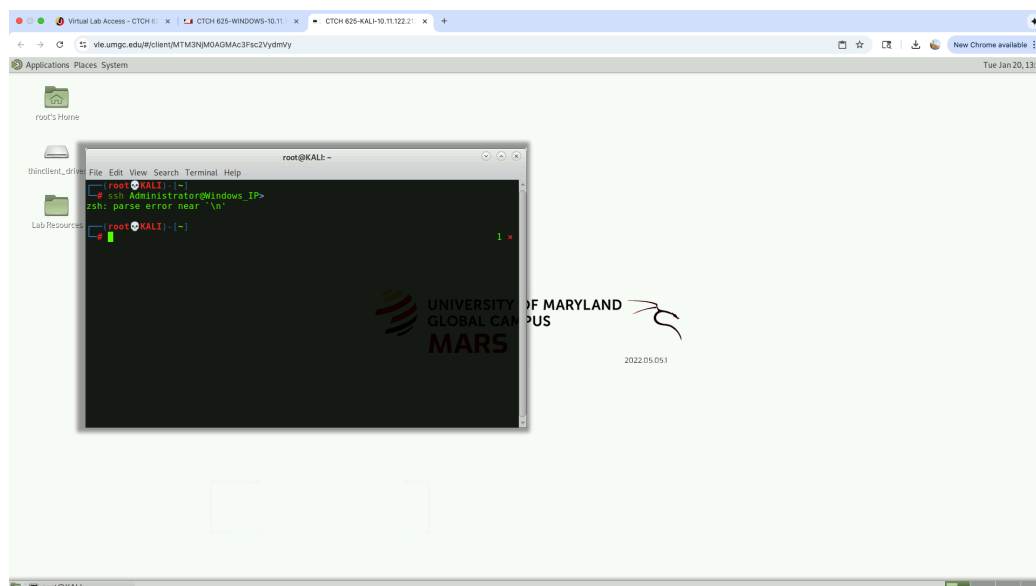


Part 2: Test SSH from Kali Windows

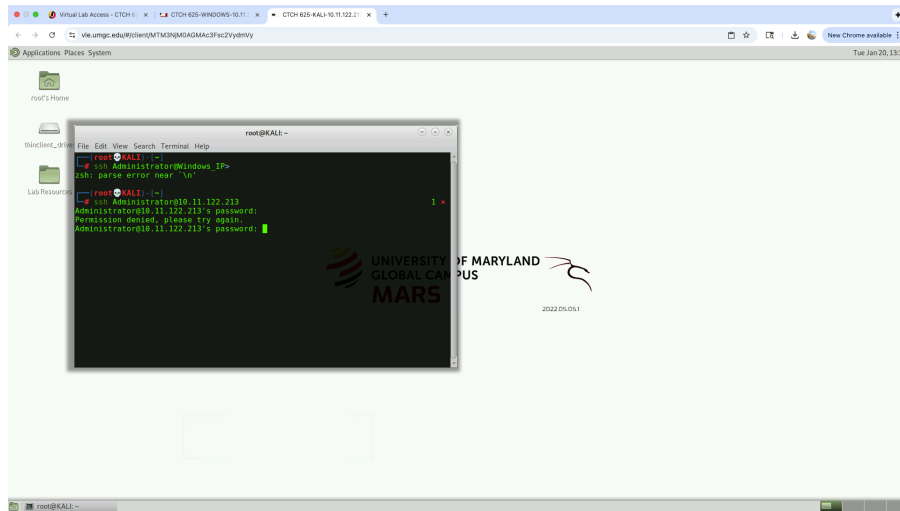
Test SSH from Kali Windows Results:

During Part 2 of the lab, I tested SSH access from the Kali Linux system to the Windows virtual machine using both the Administrator and Student1 accounts. The SSH connection successfully reached the Windows system on port 22, which confirms that the OpenSSH service was running and that the firewall rule was allowing traffic from the Kali IP address. However, all login attempts resulted in a “permission denied” or connection reset response. This behavior indicates that the issue was related to authentication rather than network connectivity. To confirm that no remote session was established, I ran the whoami command from the Kali terminal. The output showed that I was still operating as the local root user, which confirms that SSH access to the Windows system was not granted. These results demonstrate that the firewall and SSH service were functioning as expected while access controls successfully restricted unauthorized logins.

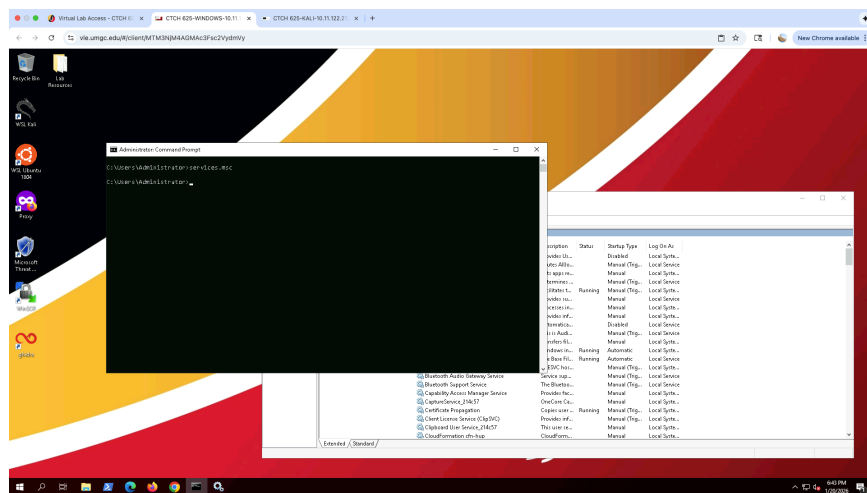
Screenshot 1: Accessing the Matte Terminal on Kali



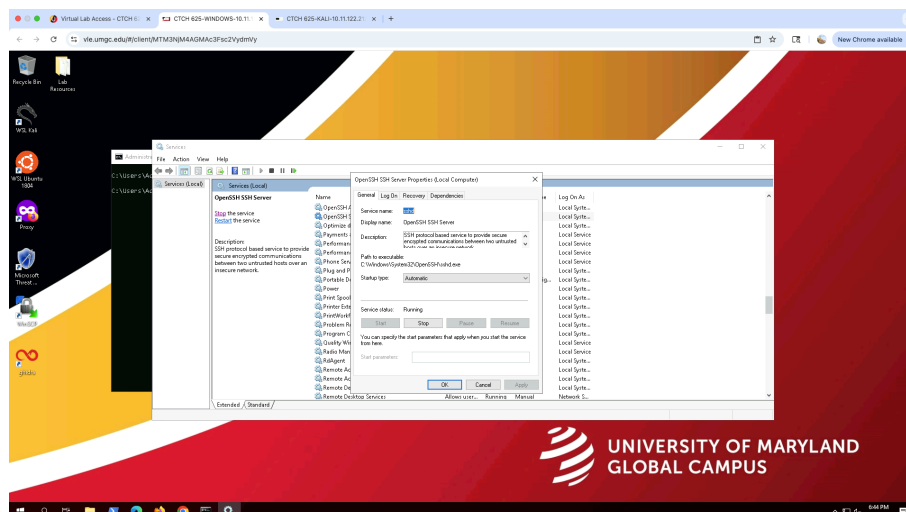
Screenshot 2: SSH Login Attempt from Kali to Window shows permission denied



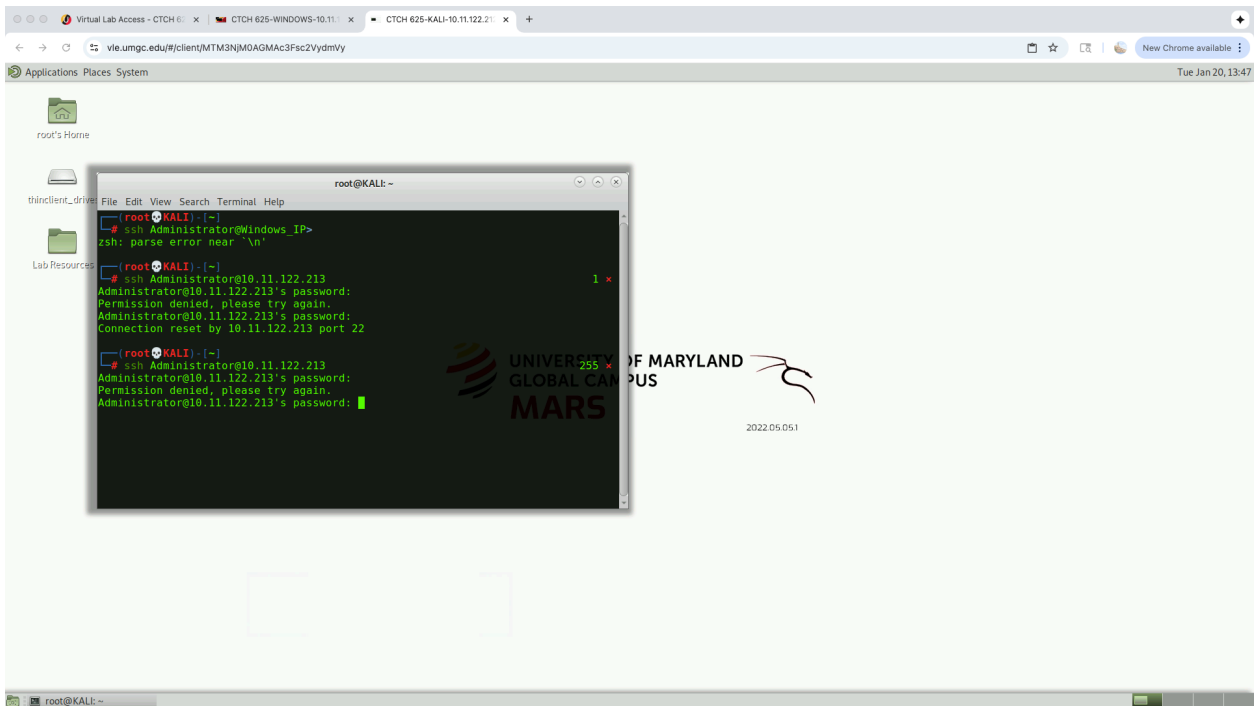
Screenshot 3: Verifying Open SSH Service Status on Windows



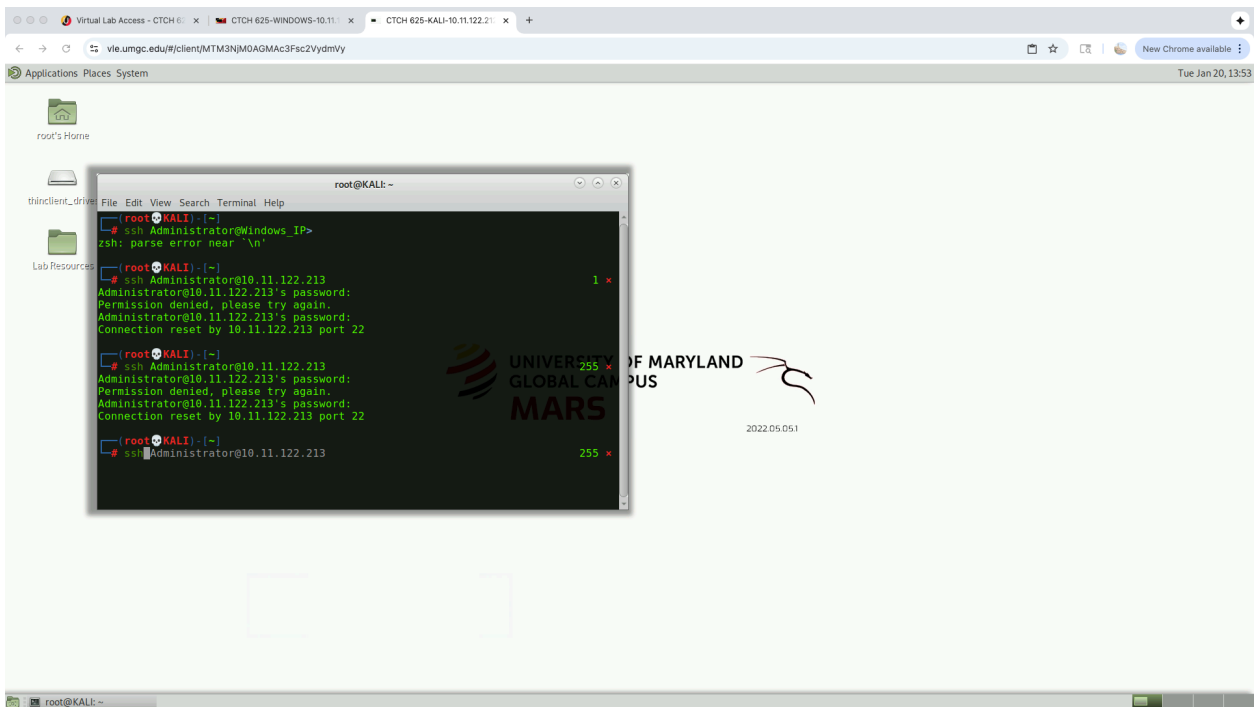
Screenshot 4: OpenSSH SSH Server Properties -Service Running



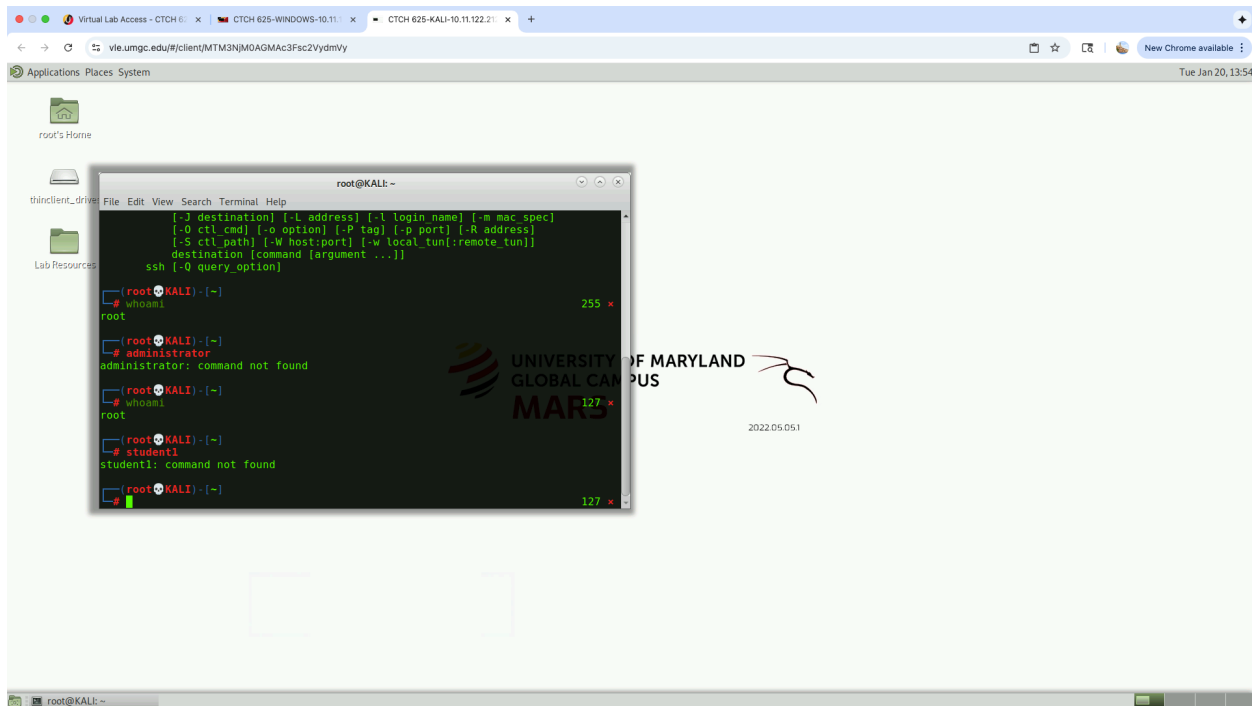
Screenshot 5: SSH Access Attempt from Kali to Window- Permission denied



Screenshot 6: Repeated SSH Authentication Failures from Kali



Screenshot 7: Kali Linux User Connect Verification Using Whomi



Part 3: Deny SSH Access by Changing Firewall Scope

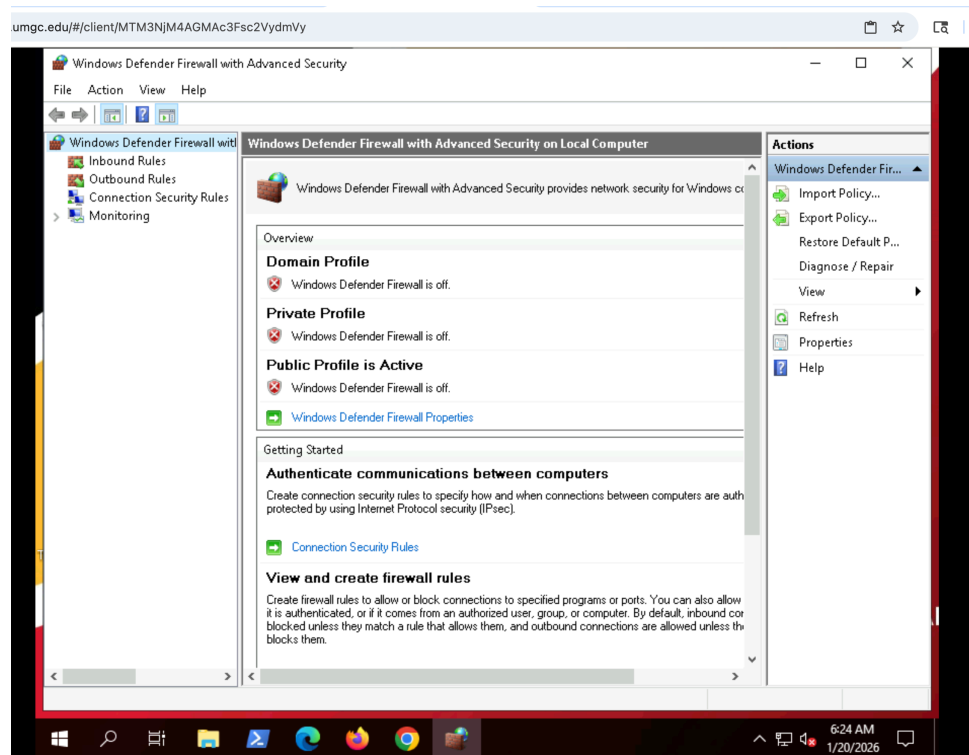
Deny SSH Access by Changing Firewall Scope Results:

For Part 3: Deny SSH Access, the OpenSSH inbound rule was modified in Windows Defender Firewall with Advanced Security. The existing OpenSSH SSH Server (Port 22) rule was updated by changing the remote IP scope. Kali's original IP address (10.11.122.213) was removed and replaced with a dummy IP address (10.10.10.10). This change prevented SSH traffic from the Kali system from being allowed through the firewall.

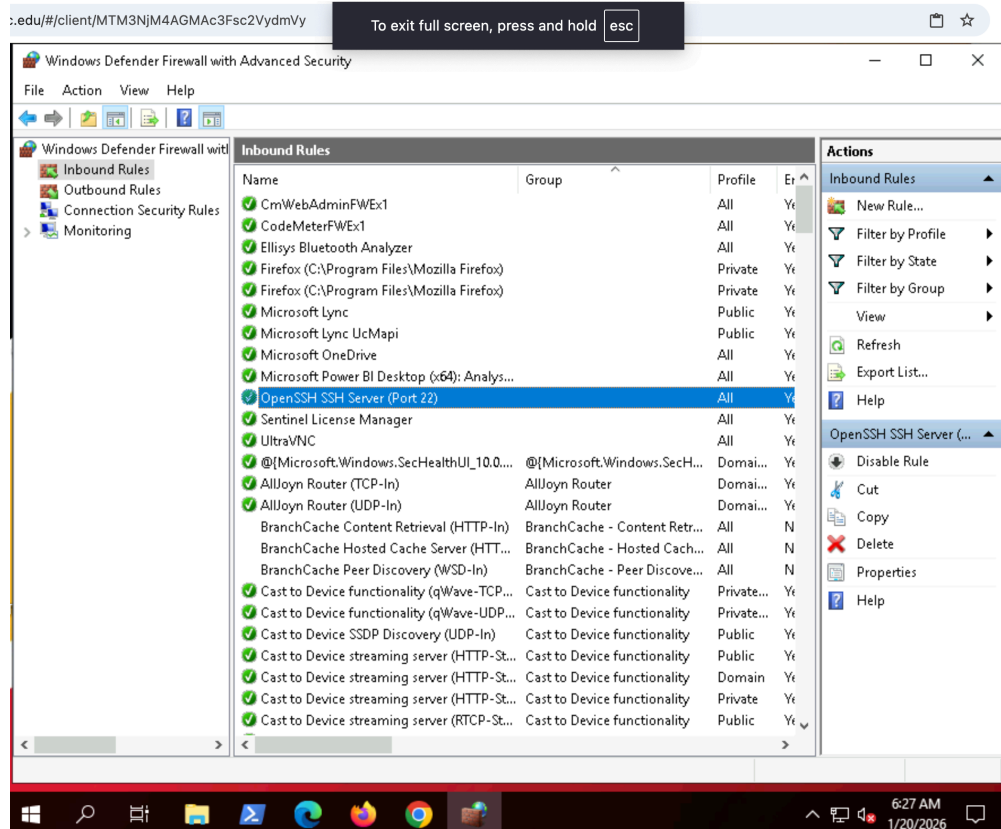
After applying the updated firewall rule, SSH connection attempts were tested from the Kali system using both the student and administrator accounts. When running the commands `ssh student1@<Windows-IP>` and `ssh Administrator@<Windows-IP>`, the SSH connection was not established. This confirms that the firewall rule successfully denied SSH access once Kali's IP address was no longer included in the allowed remote scope.

These results demonstrate how adjusting firewall scope settings can immediately control remote access. By limiting SSH access to a specific IP address, the Windows system blocks unauthorized SSH connections. This configuration supports FIC Bank's security requirements by reducing exposure to unauthorized remote access attempts and aligns with best practices for securing administrative services.

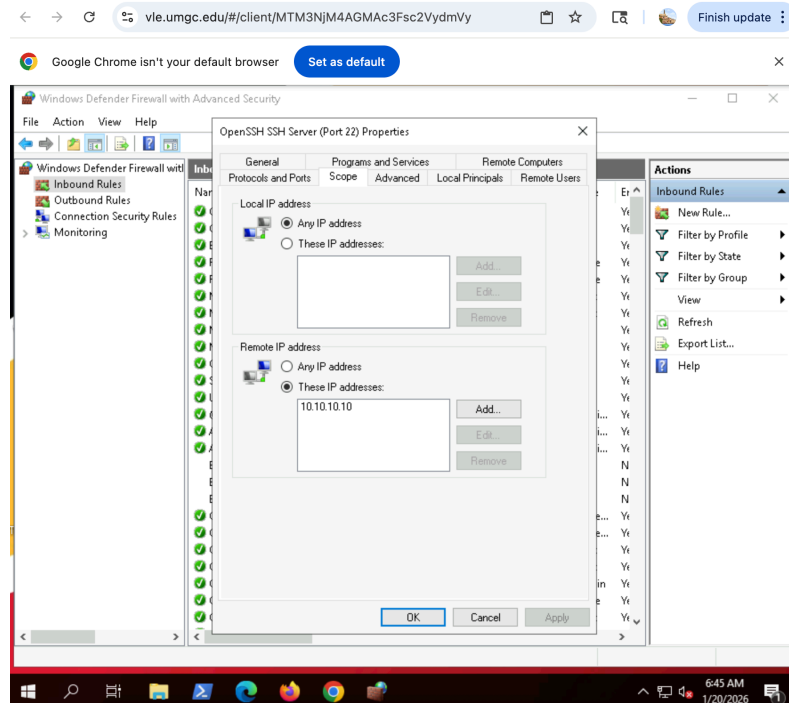
Screenshot 1: Windows Defender Firewall with Advanced Security



Screenshot 2: Inbound Rule / Open SSH Server (Port 22)

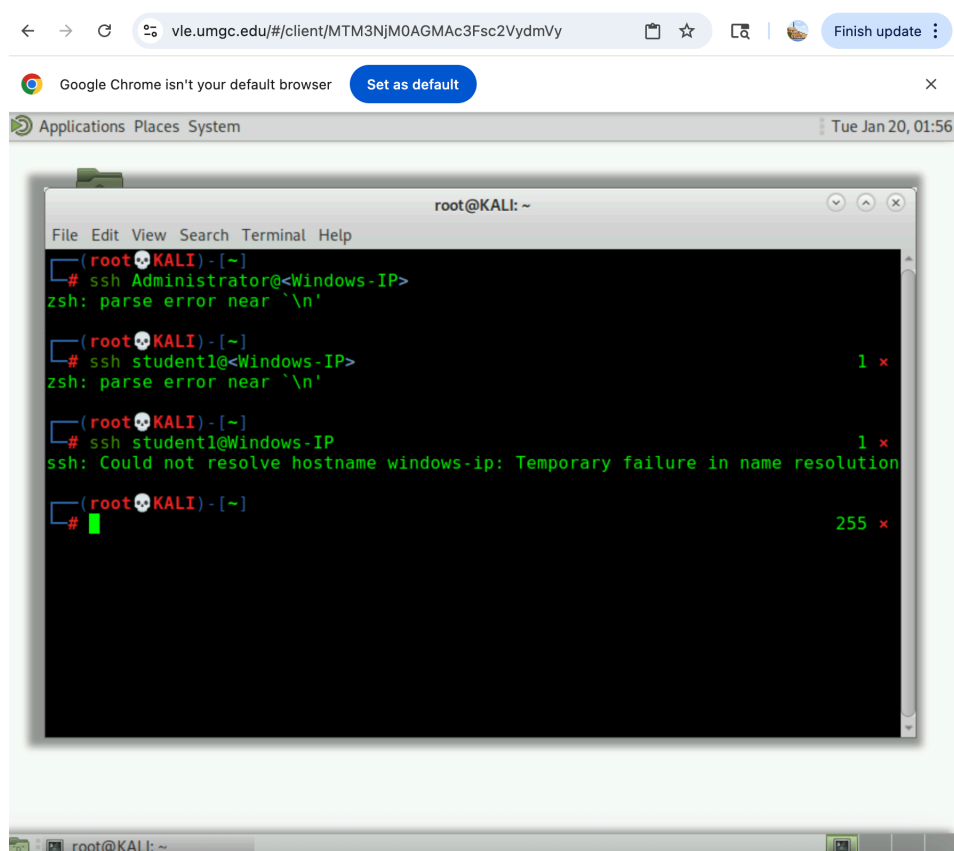


Screenshot 3: SSH Server (Port 22) Updated IP Address



Screenshot 4: Tested SSH from Kali - Used commands

ssh student1@<Windows-IP> and ssh Administrator@<Windows-IP>



Part 4: SSH from Windows to Kali

SSH from Windows to Kali Results:

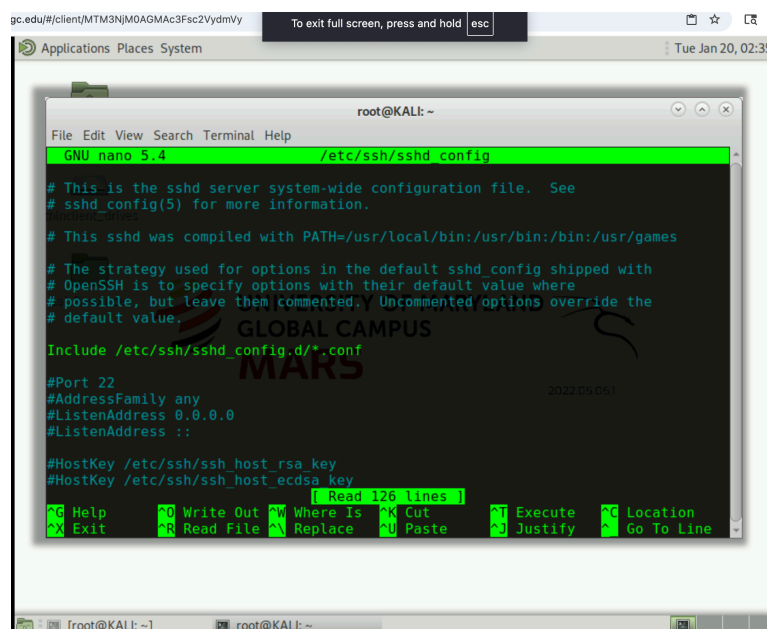
During Part 4 of the SSH from Windows to Kali exercise, Secure Shell(SSH) access from the Windows machine to Kali system was successfully configured and validated. On the kali system, the SSH server configuration file (/etc/ssh/sshd_config) was reviewed and updated to allow remote administrative access. The PermitRootLogin and PasswordAuthentication settings were enabled, ensuring that the root account could authenticate using a password-based method. After saving the configuration changes, the SSH service was restarted to apply the updated settings.

To further support secure remote connectivity, the Kali firewall was configured to allow SSH traffic. The ufw allow ssh command was executed, followed by a firewall reload to ensure the rule was enforced. These steps ensured that SSH traffic on port 22 was permitted through the Kali firewall while maintaining controlled access.

From the Windows system, Windows PowerShell was used to initiate an SSH connection to the Kali system using the verified Kali IP address. When prompted, the SSH host fingerprint was accepted, and the correct root password was entered. The connection was successfully established, and the whoami command confirmed that the session was running with root-level privileges. This outcome demonstrates that SSH access was correctly enabled, authenticated, and controlled between the Windows and Kali systems.

Overall, the results confirm that secure remote access was implemented correctly using SSH, firewall controls, and authentication mechanisms. This configuration aligns with best practices for administrative access by ensuring encrypted communication, explicit firewall permissions, and verification of user identity, supporting FIC Bank's requirement for secure remote system management in a financial environment.

Screenshot 1: SSH Configuration file (Kali) using command - sudo nano /etc/ssh/sshd_config



The screenshot shows a terminal window titled 'root@KALI: ~' with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal is running the 'GNU nano 5.4' editor, editing the file '/etc/ssh/sshd_config'. The configuration file content is displayed in a dark background with green text. The visible lines are:

```
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/bin:/usr/games

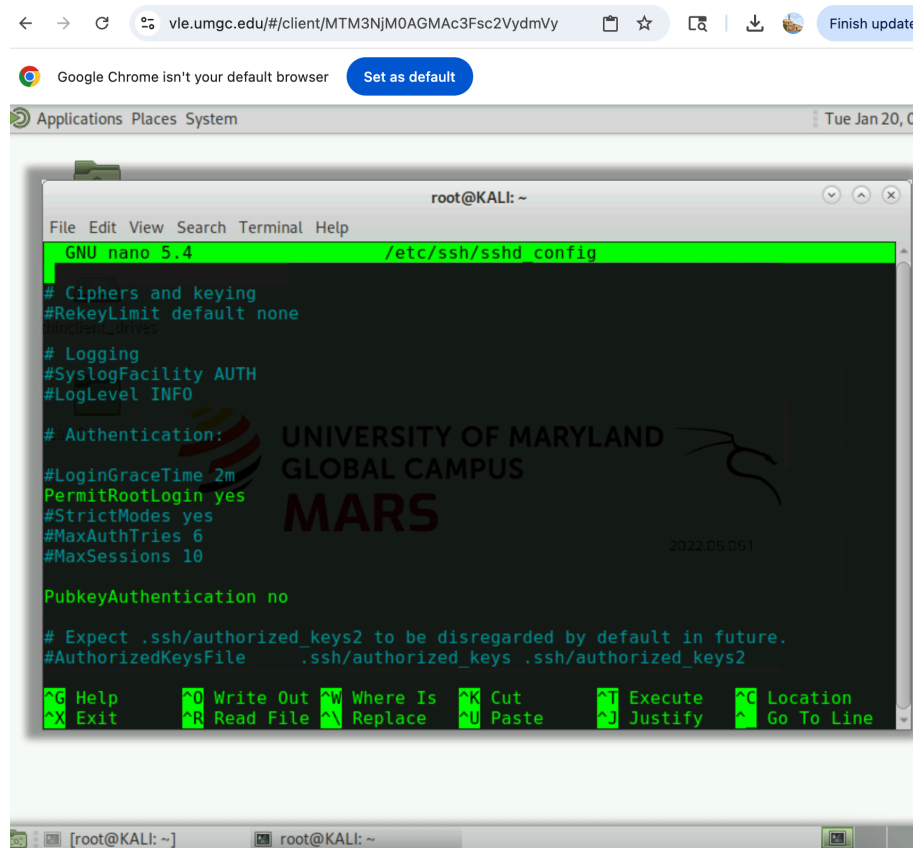
# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.
Include /etc/ssh/sshd_config.d/*.conf

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
```

At the bottom of the terminal, there is a status bar showing '[root@KALI: ~]' and a prompt 'root@KALI: ~'.

Screenshot 2: PermitRootLogin Updated



The screenshot shows a web browser window with the URL `vle.umgc.edu/#/client/MTM3NjM0AGMAG3Fsc2VydmdVY`. Below the browser, a terminal window titled `root@KALI: ~` is open, displaying the `/etc/ssh/sshd_config` file in the `GNU nano 5.4` editor. The file content is as follows:

```
GNU nano 5.4 /etc/ssh/sshd_config
# Ciphers and keying
#RekeyLimit default none
# Logging
#SyslogFacility AUTH
#LogLevel INFO

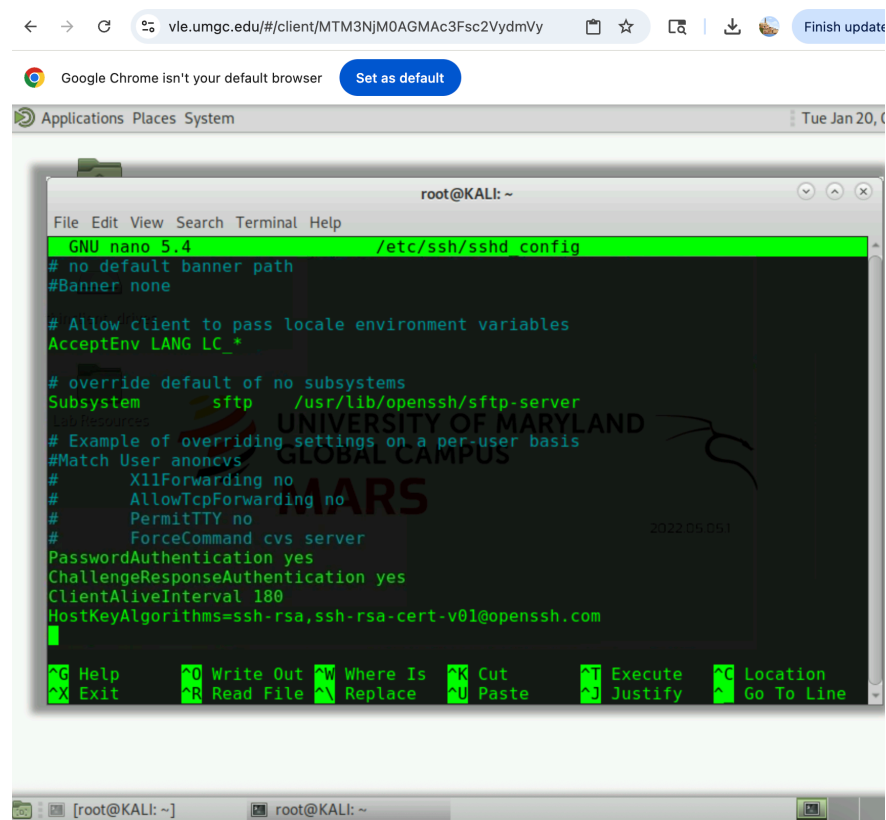
# Authentication:
#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

PubkeyAuthentication no

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2
```

The `PermitRootLogin yes` line is highlighted in green. The terminal window also displays a watermark for the University of Maryland Global Campus MARS.

Screenshot 2: Password Authentication Update



The screenshot shows the same web browser window as the previous one. The terminal window titled `root@KALI: ~` is open, displaying the `/etc/ssh/sshd_config` file in the `GNU nano 5.4` editor. The file content is as follows:

```
GNU nano 5.4 /etc/ssh/sshd_config
# no default banner path
#Banner none

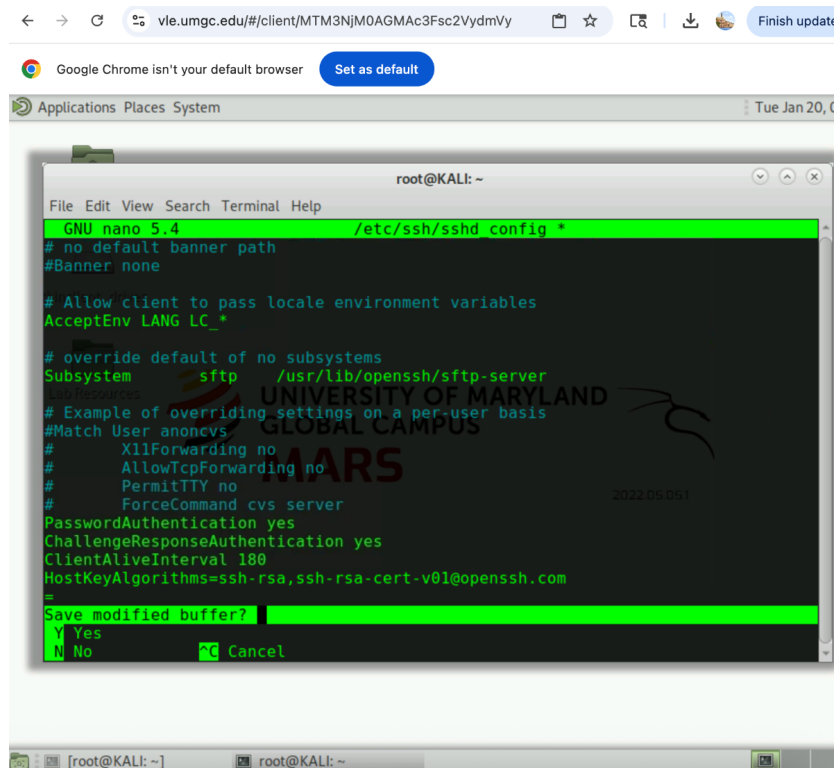
# Allow client to pass locale environment variables
AcceptEnv LANG LC_*

# override default of no subsystems
Subsystem sftp /usr/lib/openssh/sftp-server

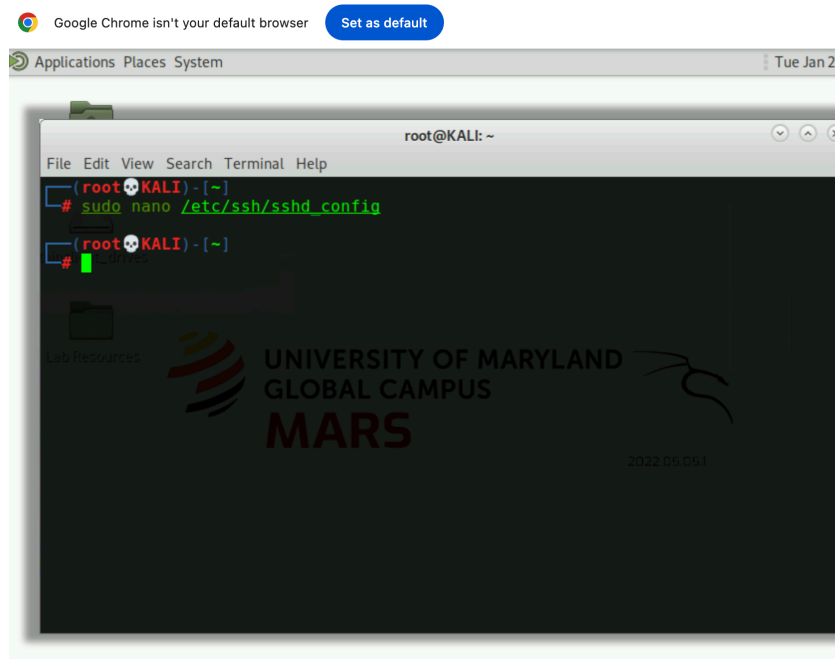
# Example of overriding settings on a per-user basis
#Match User anoncvs
#    X11Forwarding no
#    AllowTcpForwarding no
#    PermitTTY no
#    ForceCommand cvs server
PasswordAuthentication yes
ChallengeResponseAuthentication yes
ClientAliveInterval 180
HostKeyAlgorithms=ssh-rsa,ssh-rsa-cert-v01@openssh.com
```

The `PasswordAuthentication yes` line is highlighted in green. The terminal window also displays a watermark for the University of Maryland Global Campus MARS.

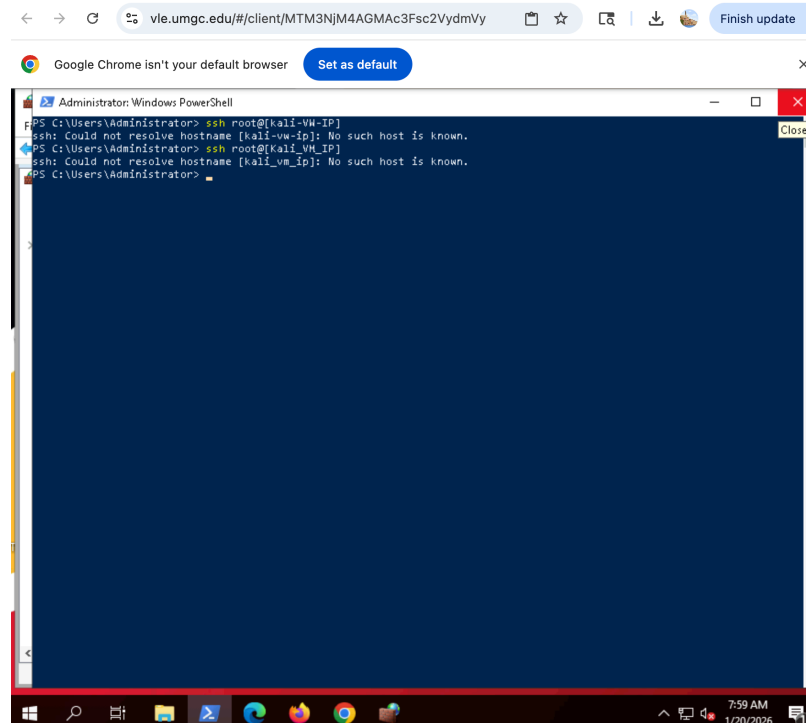
Screenshot 3: Saved Updates



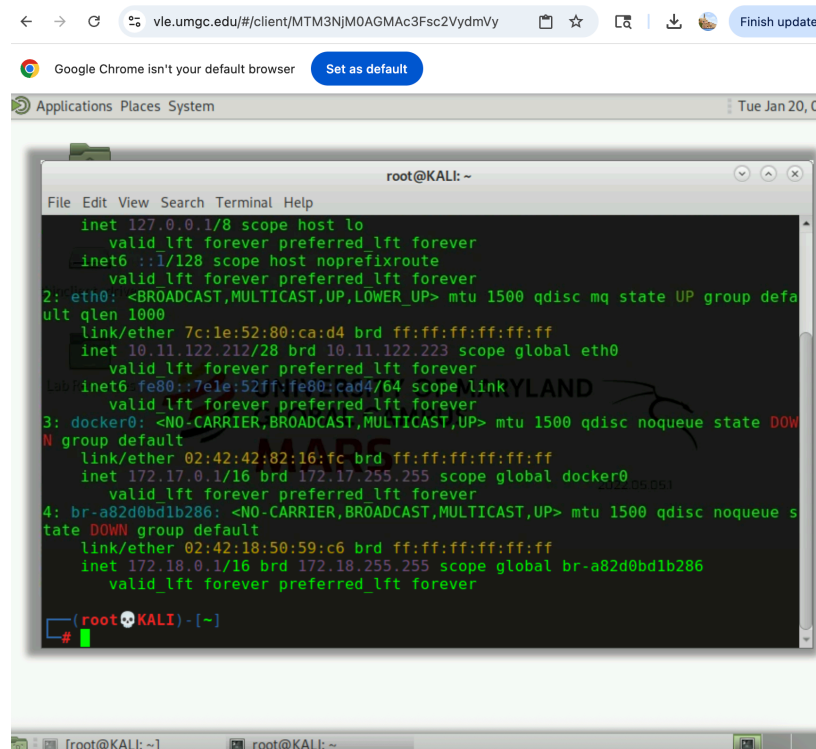
Screenshot 3: Returned Back to Root@Kali



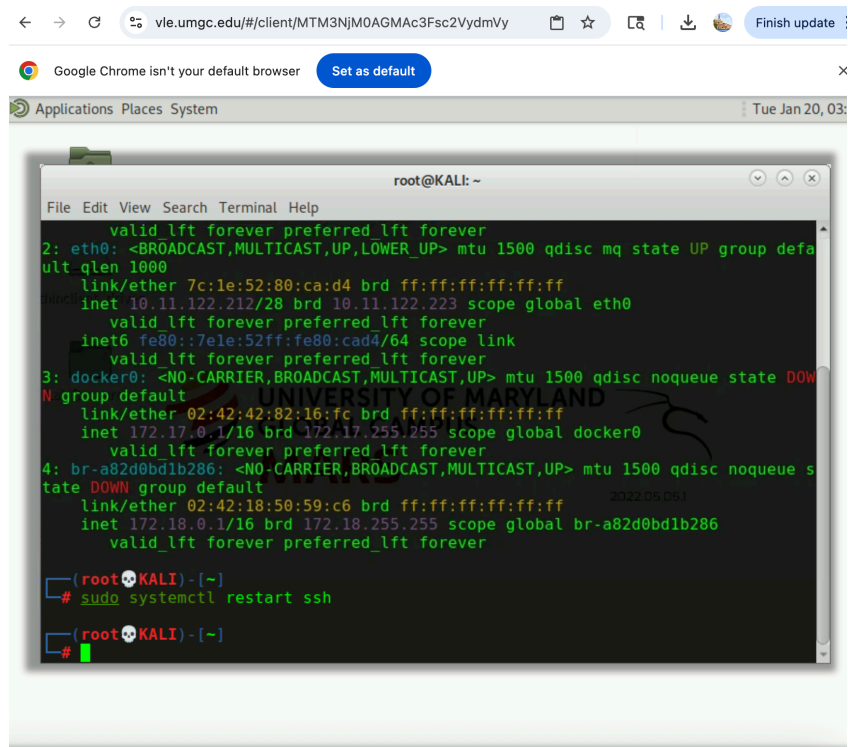
Screenshot 4: Windows PowerShell SSH attempt



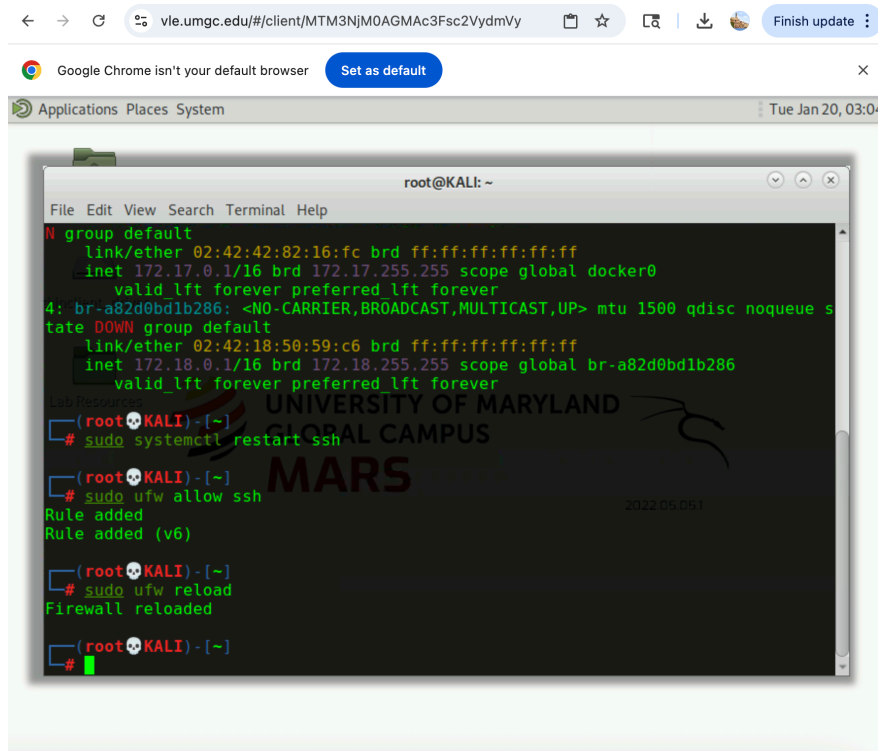
Screenshot 6: IP a output confirming Kali IP



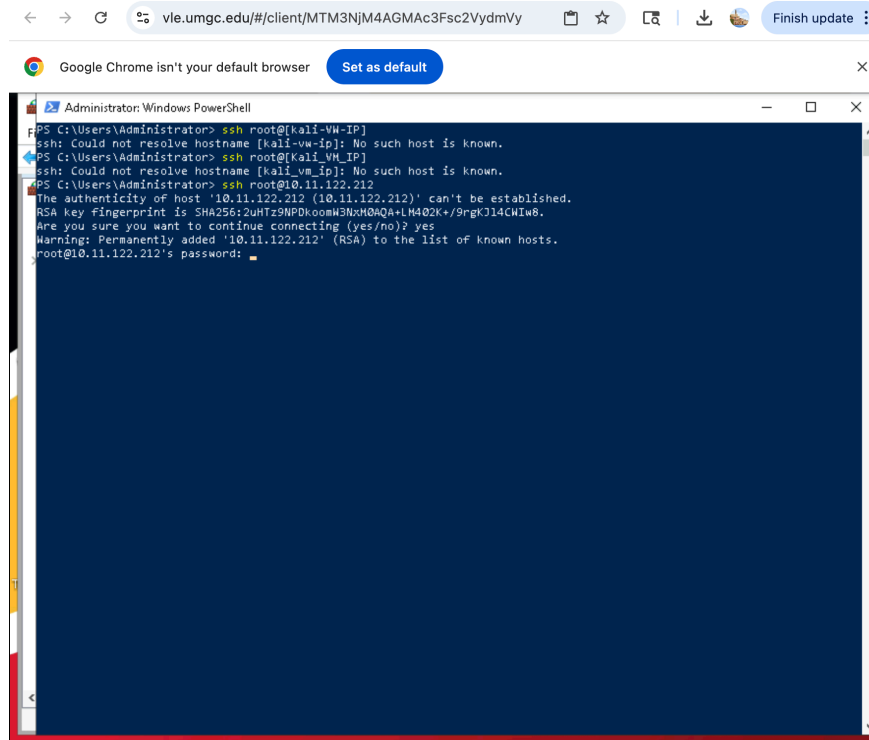
Screenshot 7: Restarting the SSH service on the Kali VM to apply configuration changes



Screenshot 8: Restarting the SSH service and allowing SSH through Kali Firewall

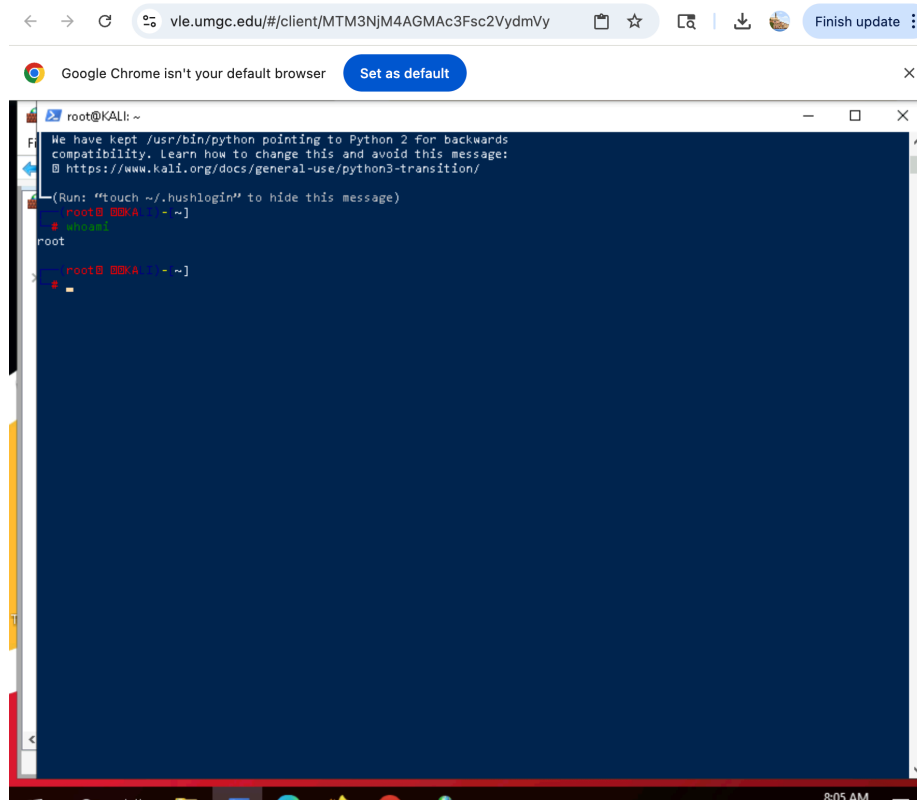


Screenshot 9: Initiating SSH connection from Windows PowerShell to the Kali VM



```
Administrator: Windows PowerShell
PS C:\Users\Administrator> ssh root@[kali-vm-ip]
ssh: Could not resolve hostname [kali-vm-ip]: No such host is known.
PS C:\Users\Administrator> ssh root@[kali-vm-ip]
ssh: Could not resolve hostname [kali-vm-ip]: No such host is known.
PS C:\Users\Administrator> ssh root@10.11.122.212
The authenticity of host '10.11.122.212 (10.11.122.212)' can't be established.
RSA key fingerprint is SHA256:2UHz9NPDkoomW3NxH0AQALM4Q2K+/9rgKJ14CHIw8.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.11.122.212' (RSA) to the list of known hosts.
root@10.11.122.212's password: 
```

Screenshot 10: Successful SSH connection verification using Whoami



```
root@KALI: ~
We have kept /usr/bin/python pointing to Python 2 for backwards
compatibility. Learn how to change this and avoid this message:
@ https://www.kali.org/docs/general-use/python3-transition/
(Run: "touch ~/.hushlogin" to hide this message)
root@kali: ~]
# whoami
root
root@kali: ~]
# 
```

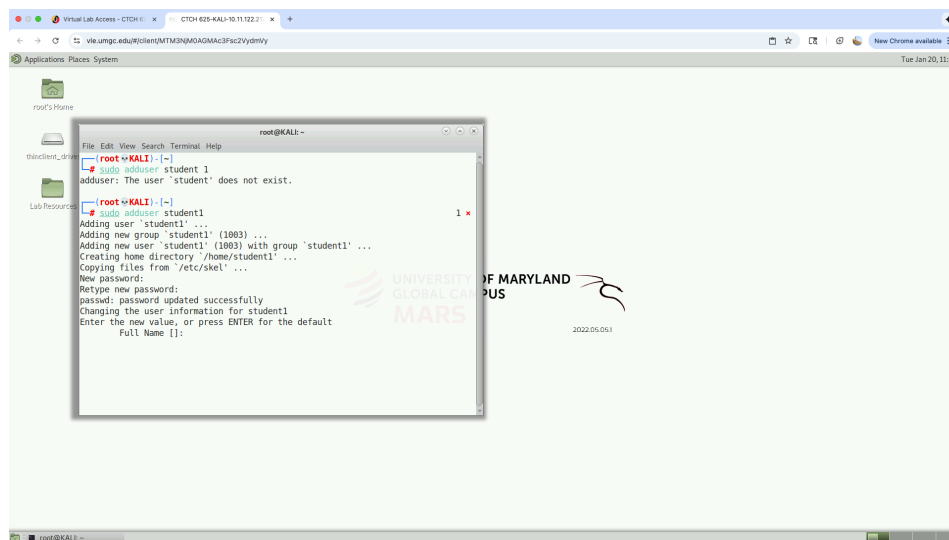
Part 5: Add a New Linux User and Test

Add a New Linux User and Test SSH Access Results:

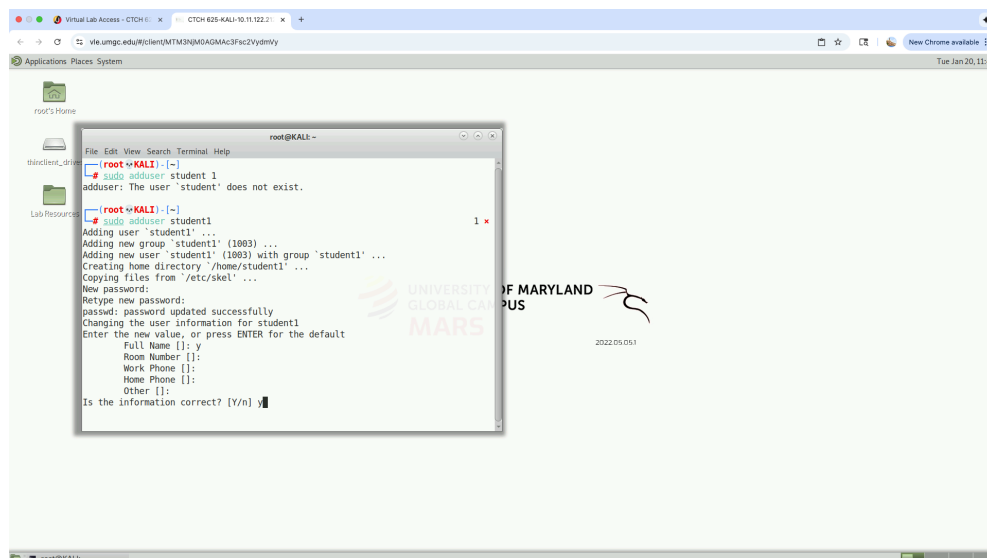
In this part of the exercise, a new Linux user account was successfully created on the Kali system to validate user-based SSH access. Using the `sudo adduser student1` command, the `student1` account was added and assigned the required password. The user creation process completed successfully, including the creation of a home directory and default user settings. After the account was created, the SSH service was restarted to ensure the system recognized the new user for remote access.

From the Windows system, an SSH connection attempt was made using the `student1` credentials and the assigned Kali virtual machine IP address. This confirmed that the newly created user account was able to authenticate remotely through SSH. These results show that SSH access on the Kali system is not limited to the root account and that additional users can be created and tested for remote access, demonstrating proper user management and controlled SSH connectivity.

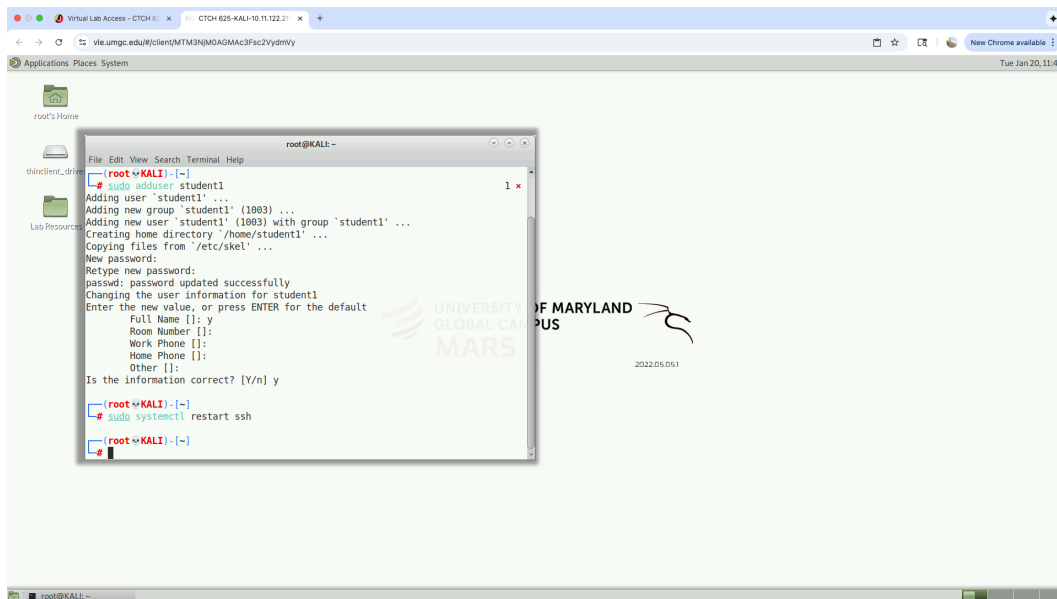
Screenshot 1: Add New User on Kali Linux VM



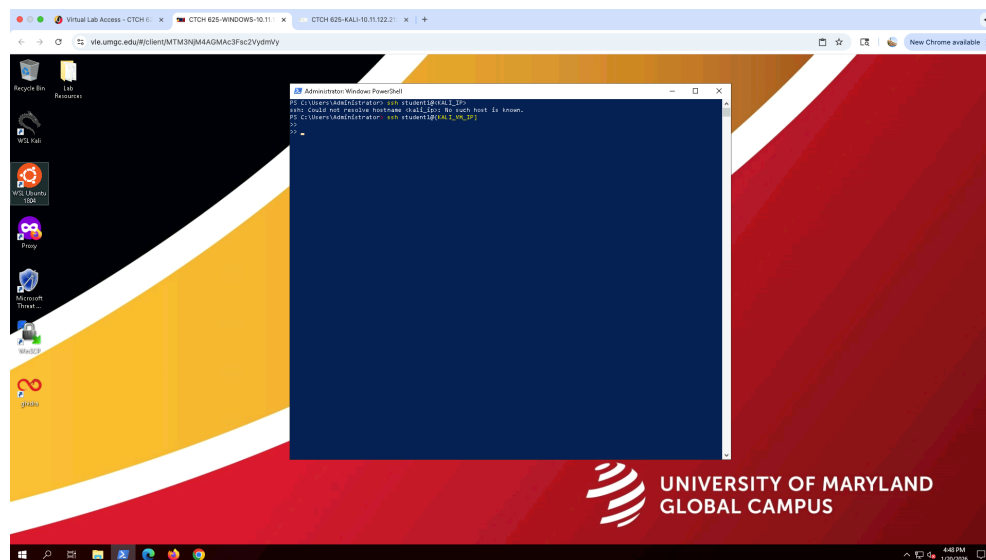
Screenshot 2: Confirm User Details and Complete Account Creation



Screenshot 3: Restart the SSH Service on Kail



Screenshot 4: Attempt SSH login as the New User from Windows/ SSH Test with New Linx User



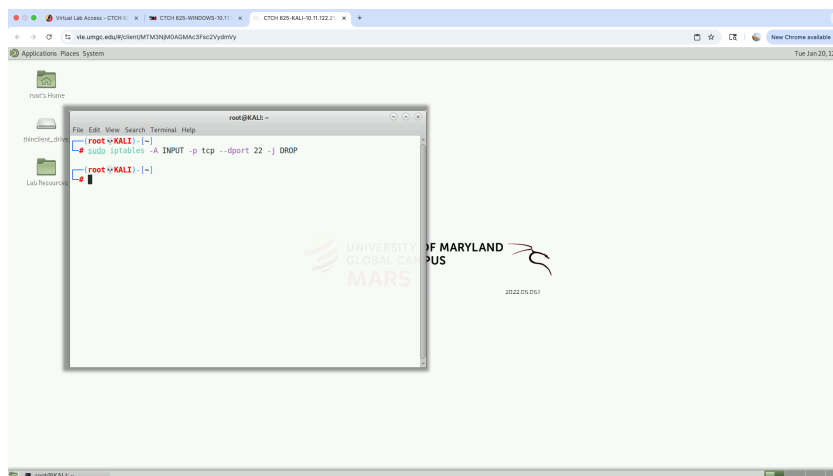
Part 6: Test Linx Firewall

Test Linx Firewall Results:

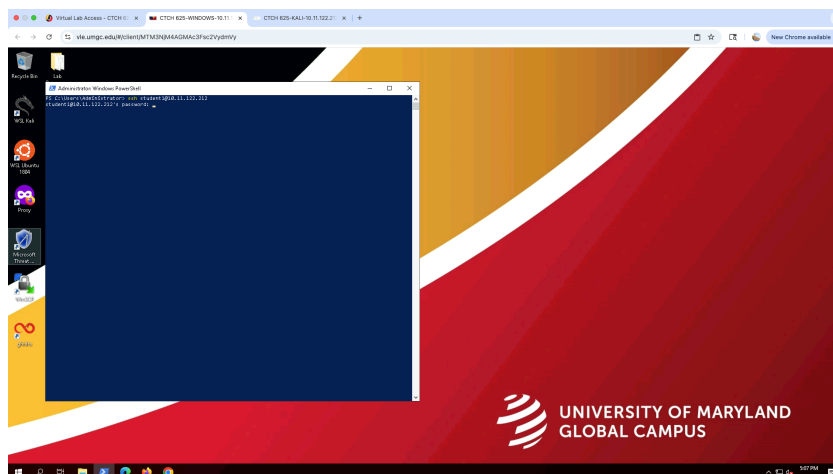
In this part of the exercise, the Linux firewall was tested on the Kali system using IPTables to control SSH access. An IPTables rule was added to block incoming SSH connections on port 22. After the rule was applied, an SSH connection attempt from the Windows system failed, confirming that the firewall was actively preventing remote SSH access. This demonstrated that IPTables can immediately restrict network traffic when a rule is enforced.

Once the firewall rule was removed from the INPUT chain, SSH access was tested again from the Windows system. This time, the connection was successful, showing that SSH access was restored after the rule was deleted. These results confirm that IPTables can be used to dynamically block and re-enable SSH access on a Linux system, reinforcing how firewall rules directly impact network connectivity and support secure access control.

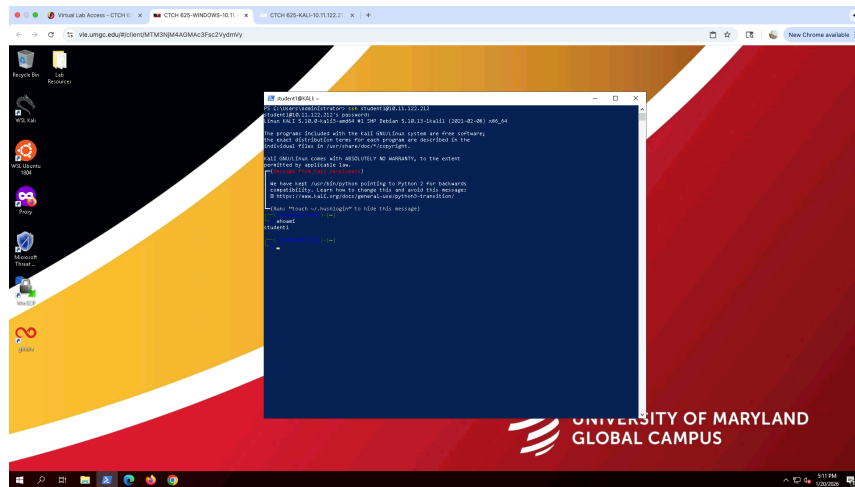
Screenshot 1: Block SSH on Kali



Screenshot 2: Successful SSH Login from Windows to Kali using student 1 account



Screenshot 3: Verification of SSH Access Using the Newly Created Student1 Account



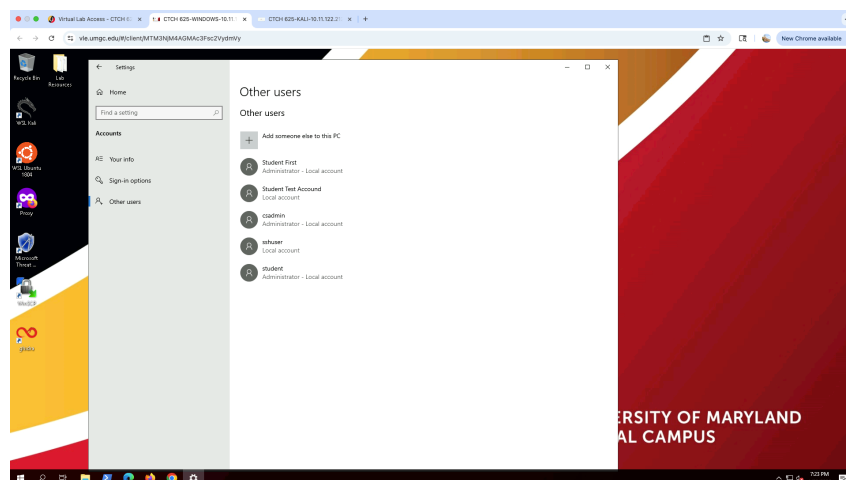
Part 2: Windows Firewall and Access Restrictions

Student2 password Update Results:

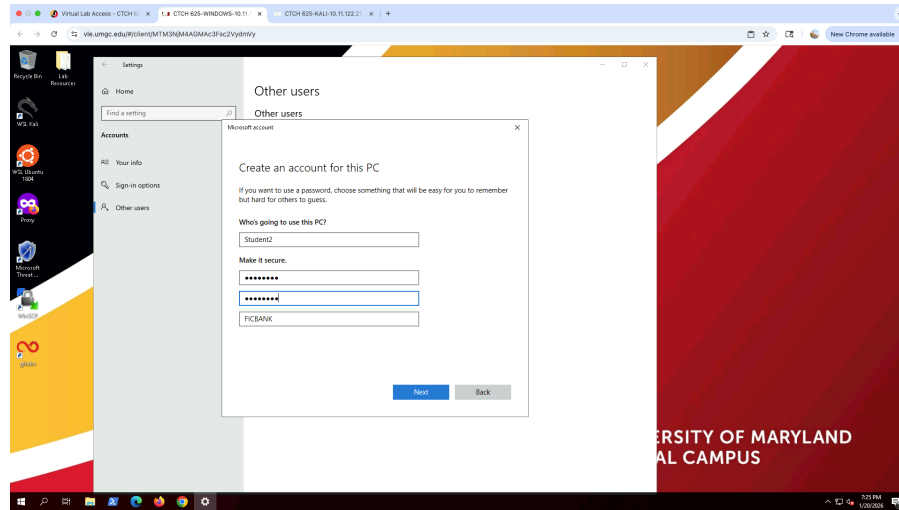
During this task, I created a new local user account named Student2 and assigned an initial password that met the required complexity guidelines. After the account was created, I attempted to log out of the Administrator account in order to log in as Student2. However, each time I signed out, the virtual lab environment automatically redirected the session back to the Administrator account. This behavior appeared to be a limitation of the lab setup rather than an error in the configuration.

Since I was unable to gain access as Student2, I proceeded with the password update using the available security options. I accessed the Ctrl + Alt + Del menu and selected the Change a password option, where I updated the account password to a stronger value that met all required criteria, including length, uppercase and lowercase letters, numbers, and special characters. The system confirmed that the password was successfully changed. Overall, this process demonstrates how user accounts can be created and secured even when interactive login is restricted, and it highlights the importance of strong password enforcement to reduce the risk of unauthorized access.

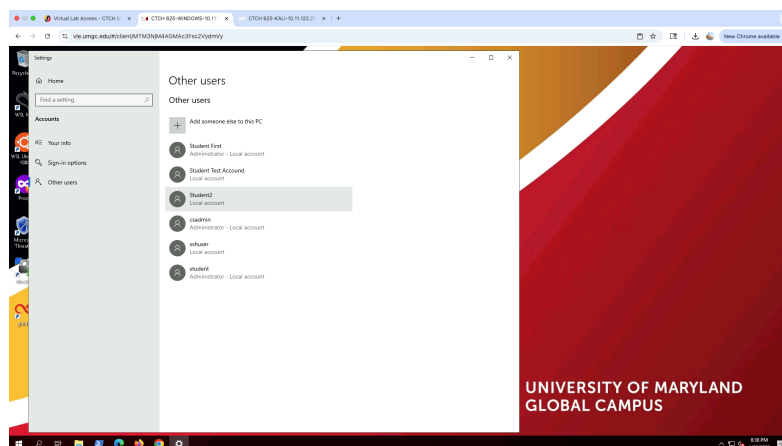
Screenshot 1: Navigating to Other Users



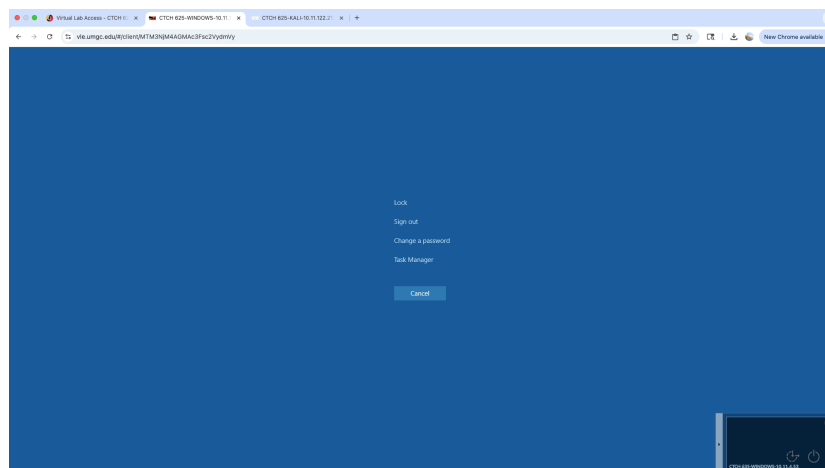
Screenshot 2: Confirmation of Student 2 Account Creation



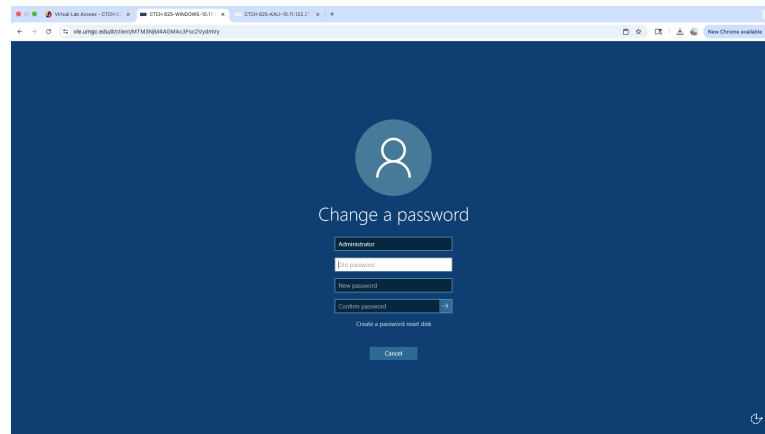
Screenshot 2: Confirmed Student2 is added



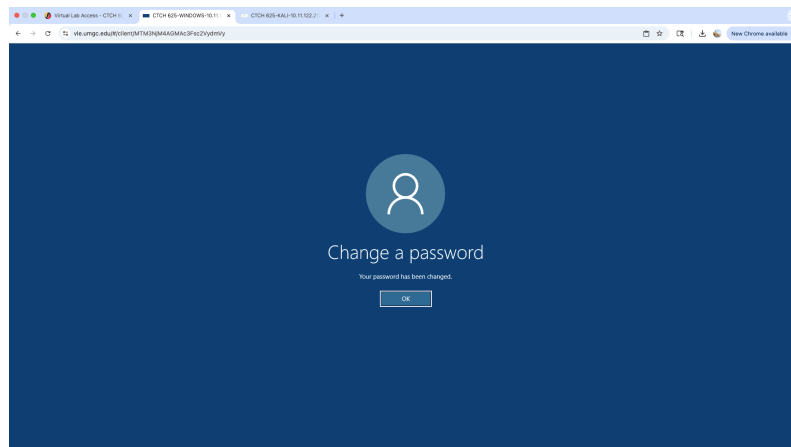
Screenshot 3: Attempted to logout - Failed



Screenshot 3: Changed Student2 Password- Username was Changed to Student2



Screenshot 3: Student2 Password-Updated!



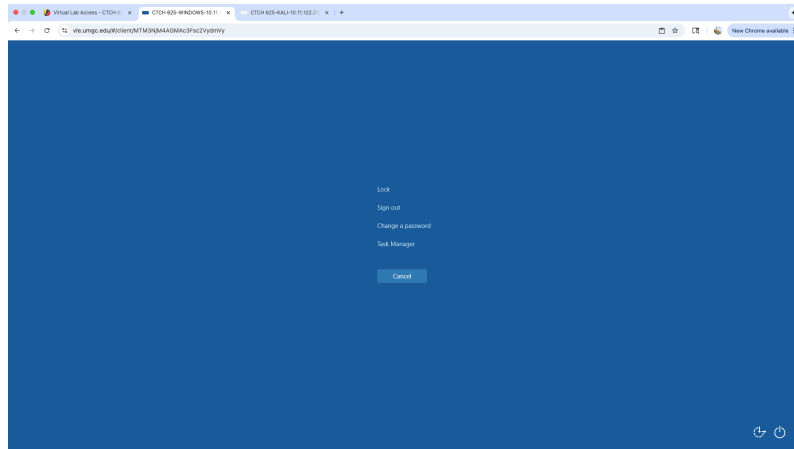
Part 3: Test New Student2 Password

New Password test results:

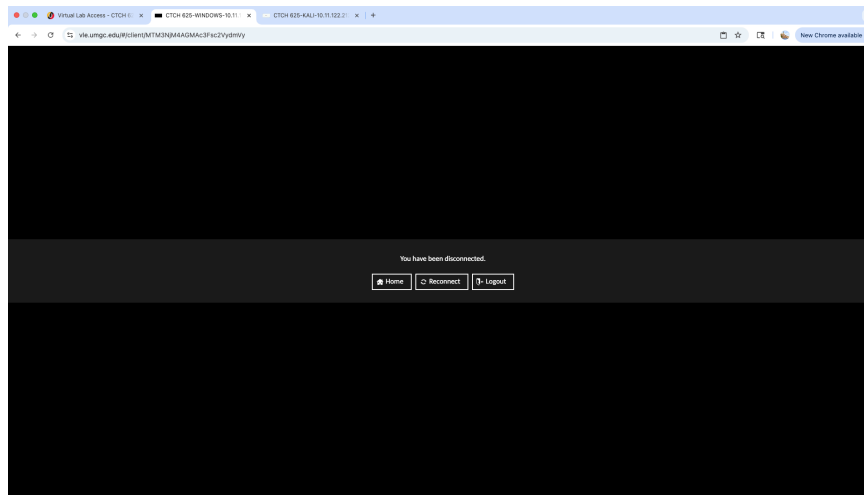
In this task, I attempted to sign out of the Administrator account as instructed; however, the system did not allow a normal sign-out process. Each time the Sign out option was selected, the session redirected back to the Administrator screen instead of completing a sign-out that would allow access to the Student2 account. As a result, I was never able to fully sign in to the Student2 account or reach a separate Student2 login screen.

Despite this limitation, I confirmed that the Student2 user account was successfully created and that the username update and password configuration were applied correctly. Screenshots were captured to document the sign-out attempt and the system's redirection behavior as proof of results. These findings show that while the account was properly created and managed, the lab environment restricted user sign-out and prevented full login testing for the Student2 account.

Screenshot 1: Log Out of the Current Account



Screenshot 2: Sign out Results



2. FIC Bank Network Security Needs:

FIC Bank's network security needs require limiting exposure and validating access in order to maintain remote access control best practices for RDP and SSH. Due to the nature of the organization, it is critical to implement security controls rather than allowing unrestricted connectivity. As a global financial institution handling sensitive data, FIC Bank must treat RDP and SSH as high-risk services, especially during aggressive inbound and outbound network scanning. To align with this policy, access was intentionally limited through firewall rules and access restrictions so that only authorized connections were allowed. This approach reduces the attack surface while supporting early detection of suspicious activity. Although cyberattacks cannot be completely prevented, controlling remote access and monitoring traffic helps identify security gaps and lowers the risk of compromise, which is essential for protecting FIC Bank's systems and global network operations (Wolkstein, 2025).

During the hands-on project, I implemented and tested firewall and access control configurations in alignment with FIC Bank's security requirements. This included creating and managing user accounts, modifying Windows Firewall rules, and validating how access was permitted or denied under different conditions. Firewall rules were used to restrict inbound RDP and SSH traffic so that these services were not accessible to all remote connections. SSH access was tested using both authorized and unauthorized user accounts to observe how the system responded to access attempts. When access was denied, the results confirmed that the firewall and authentication controls were functioning as intended rather than allowing unrestricted connectivity. These steps align with FIC Bank's aggressive scanning policy by reducing the attack surface and making unauthorized connection attempts more visible during monitoring. By enforcing controlled entry points and documenting both successful and denied access attempts, the configuration supports the bank's ability to detect abnormal traffic patterns and respond quickly to potential threats.

Overall, this process demonstrated how remote access best practices support both system security and continuous monitoring. By tightening access controls and relying on firewall enforcement, FIC Bank is better positioned to prevent unauthorized connections while still allowing secure administrative access when required.

3. Firewall Rules and Remote Access Security

During the hands-on exercise, several firewall rules were created and modified to control how remote connections were allowed or denied for RDP and SSH. On the Windows system, inbound firewall rules were adjusted to restrict SSH access based on specific IP addresses rather than allowing connections from any source. By specifying exact IP addresses, the firewall ensured that only approved systems could attempt to connect. This process helped reduce unnecessary exposure to external traffic. When unauthorized IP addresses attempted to connect, the firewall correctly blocked the traffic, confirming that the rule was working as intended. This approach helps secure remote connections by limiting access points and preventing attackers from accessibility of scanning or accessing remote service, which aligns with recommended remote access security practice (CISA, 2023).

On the Kali system, firewall rules were tested using IPTables to directly block and re-enable SSH traffic on port 22. Adding a rule to drop SSH traffic immediately stopped remote access attempts, while removing the rule restored connectivity. This demonstrated how firewall rules contribute to securing remote connection by enforcing strict access controls, supporting encrypted communication and making unauthorized access attempts easier to detect during monitoring.

However, configuring firewall rules for FIC Bank's global network can present several challenges. FIC Bank operates across multiple intentional locations, firewall rules must account for changing IP addresses, remote offices, and authorized third party access. Overly restrictive rules could unintentionally block legitimate users, while overly permissive rules could increase security risks. Additionally, aggressive scanning policies may generate large volumes of traffic, making it more difficult to distinguish between legitimate scanning activity and malicious behavior. These challenges highlight the importance of carefully balancing security and

accessibility while continuously reviewing and updating firewall configurations to support secure remote access across FIC Bank's global infrastructure.

4. Managing User Accounts (Corporate vs Personal Systems)

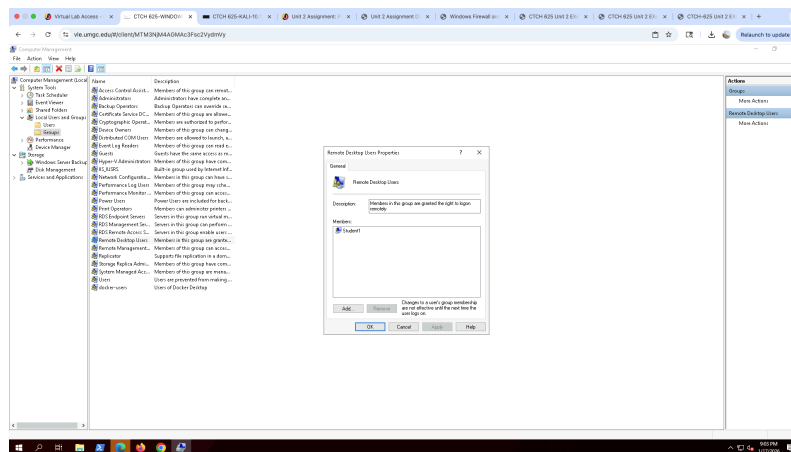
Based on hands-on experience, managing user accounts in a corporate environment is very different from managing user accounts on a personal computer due to security requirements, scale, and centralized control. In a corporate environment such as FIC Bank, user account management is structured around protecting sensitive systems and financial data rather than individual convenience. Because FIC Bank operates as a financial institution, additional security controls are required to protect the overall network. These controls include carefully configured firewalls, role-based access, and strict authorization policies. While firewalls cannot completely prevent cyberattacks, they play an important role in reducing risk and limiting exposure by controlling how users and systems access network resources. This approach follows the principle of least privilege, ensuring users are granted only the minimum level of access required to perform their job function within a centralized access control model,

Corporate environments also rely heavily on controlled remote access models. Remote desktop access may occur either directly between a remote client and an internal system or indirectly through a trusted intermediary. Direct access is often restricted by firewalls to reduce risk and enforce security policies (Souppaya, 2016). The screenshots from this project demonstrate how access is managed through group-based authorization rather than automatic permission. When the student account was added to the Remote Desktop Users group, access was allowed. Once the account was removed from the group, access was denied. This confirms that corporate systems enforce role-based access and layered security controls rather than relying on simple user creation alone. Remote access servers must remain fully patched, securely configured, and managed only by authorized administrators to maintain a strong security posture (Souppaya, 2016). This configuration demonstrates that access control decisions are enforced at the system and policy level, rather than being dependent on individual user discretion, which is critical for maintaining security in a regulated financial environment.

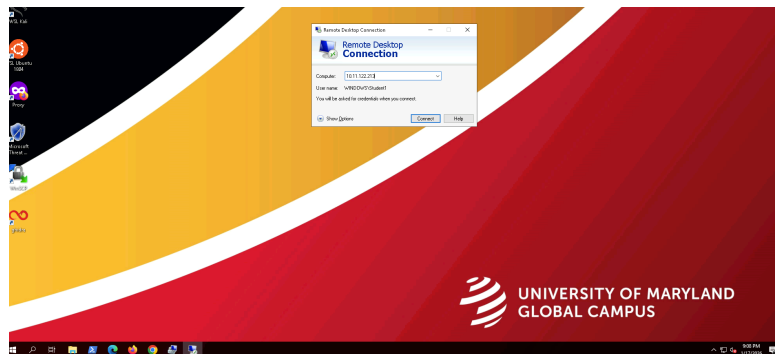
In contrast, user account management on a personal computer is much simpler and less restrictive. Personal systems typically prioritize ease of use over strict access control and often rely on basic firewall protection without centralized oversight. While personal systems may use hardware or software firewalls, they do not provide the same level of structured access management, monitoring, or policy enforcement found in corporate environments. As shown in the screenshots, FIC Bank's environment requires users to be explicitly authorized through group membership and firewall controls before remote access is permitted. This level of control is necessary in an enterprise setting to protect sensitive data and ensure that access is granted only based on job roles and business needs rather than convenience. Without centralized monitoring, enforced group membership, and formal authorization policies, personal systems introduce a higher risk of unauthorized access, misconfiguration, and delayed incident detection compared to corporate environments like FIC Bank.

6. Screenshots of Corporate environment vs personal computer

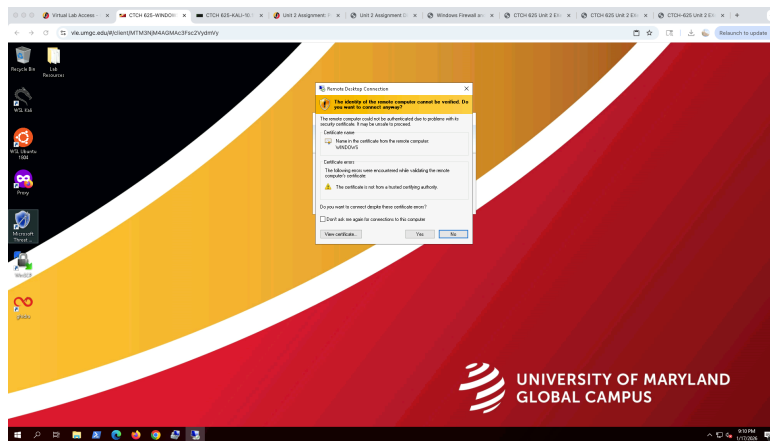
Screenshot 1: Remote Desktop user group configuration (student add)



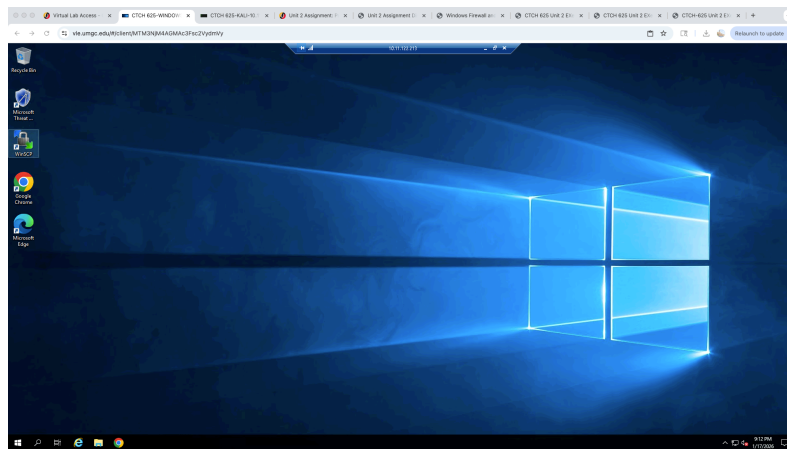
Screenshot 2: Remote Desktop Connection Attempt using student account



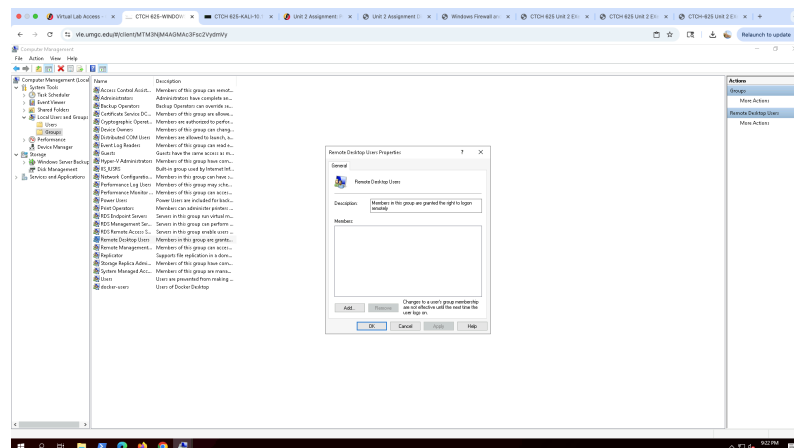
Screenshot 3: Remote Desktop Certificate Warning



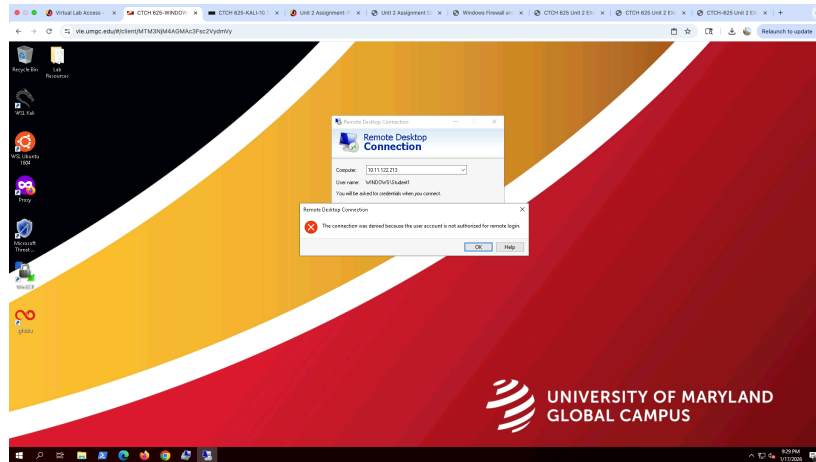
Screenshot 4: Successful Remote Desktop Login to Window System (establish)



Screenshot 5: Verification of remote desktop user group membership /remove group empty

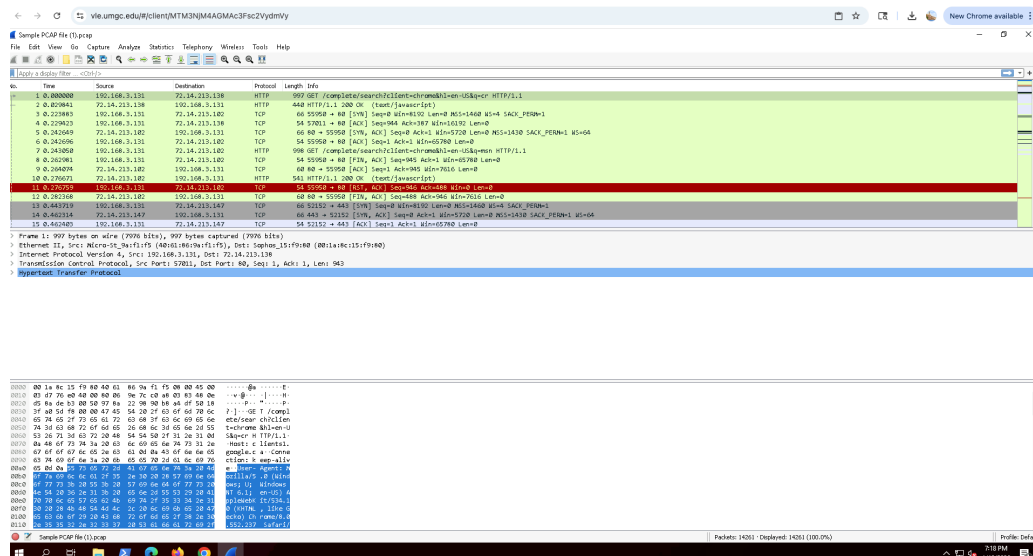


Screenshot 6: remote desktop connection attempt denied for student 1



Exercise 3: Examine secure remote access control implementation test results.

Screenshot 1: Wireshark PCAP File Results



The IP address primary source observed in the PCAP file is 192.168.3.131. This IP address appears repeatedly in the Source column and represents the internal host that is initiating most of the network communication captured in the file. The IP addresses destination is identified in the packet capture, which includes:

- 72.14.213.102
- 72.14.213.131
- 72.14.213.138
- 72.14.213.147

These IP addresses appear in the Destination column and represent external systems that the source host (192.168.3.131) was communicating with during the capture network traffic

Network Protocols:

There are many different network protocols, each designed to serve a specific purpose in network communication. Understanding the main functions of these protocols is important when analyzing network traffic. Based on the evidence observed in the PCAP file, several protocols were used to support communication between devices on the network.

One of the primary protocols identified is Transmission Control Protocol (TCP), which works alongside Internet Protocol (IP). TCP/IP is responsible for breaking data into packets, sending them across the network, and ensuring the packets arrive at the destination in the correct order. TCP also provides reliability by detecting errors and retransmitting packets when data is lost during transmission. TCP establishes trusted connections between systems, which can be targeted by attackers attempting to send malicious or fake requests that appear to come from legitimate devices (Greene, 2024). This reliability is critical for remote access protocols such as SSH and RDP, which depend on stable TCP connections to function properly.

The second protocol identified is Transport Layer Security (TLS), which provides cryptographic protection for data as it travels across networks. TLS works by encrypting network traffic and authenticating the communicating parties, which helps prevent tampering and eavesdropping during transmission (Greene, 2024). This protocol is especially important for FIC Bank because sensitive data must be protected while being sent or received. TLS also protects credentials and session data during secure remote access connections. Keeping systems updated and ensuring proper authentication helps maintain secure communication within the network.

The third protocol identified is the Domain Name System (DNS), which allows website names to be translated into numerical IP addresses that computers and network devices use to locate the correct websites and content (Greene, 2024). The evidence in the PCAP file shows DNS-related activity involving IP addresses, which can indicate either normal network communication or potential attempts to target or misuse network resources. If DNS is not properly managed and secured, FIC Bank could experience issues such as website disruptions or increased traffic caused by malicious activity, which could impact the availability of online services.

The fourth protocol identified is Hypertext Transfer Protocol (HTTP), which is used to transfer information across a network between a web browser and a web server. HTTP allows users to request and receive web content such as webpages and data files (F5, n.d.). However, HTTP traffic is not encrypted, which means information being transmitted can be exposed while

communication is in progress. According to CISA, if an attacker is able to view network traffic, the content can be intercepted or tampered with. In this packet capture, HTTP traffic is shown using port 80 and includes GET requests and responses, which demonstrates standard web communication activity. This highlights the risk of using HTTP instead of encrypted protocols when sensitive data is involved.

The last protocol identified is Hypertext Transfer Protocol Secure (HTTPS), which is the secure version of HTTP and is used to send data between a web browser and a website. HTTPS increases the security of data transfer by using encryption to protect information while it is being transmitted. According to Cloudflare, HTTPS helps prevent website information from being broadcast in a way that could be easily viewed or intercepted by attackers on the network.

Overall, the PCAP file shows how multiple network protocols work together to support communication across a network. The source IP address observed in the capture initiated communication with several destination IP addresses, demonstrating normal client-to-server interactions. Protocols such as TCP, DNS, HTTP, HTTPS, and TLS each play a specific role in enabling reliable data transfer, name resolution, and web communication. TLS and HTTPS provide important security services by encrypting data in transit and protecting sensitive information from interception or tampering. Reviewing these protocols in the PCAP file helps illustrate how secure and non-secure communications appear in network traffic and why strong encryption and proper configuration are critical for protecting FIC Bank's systems and data. This analysis also supports FIC Bank's aggressive network scanning policy by showing how encrypted and unencrypted traffic can be identified during monitoring.

Protocol Providing Security Service:

In addition to TLS and HTTPS, Secure Shell (SSH) also provides security services by encrypting remote administrative sessions. The packet exchanges, Transport Layer Security (TLS) and Hypertext Transfer Protocol Secure (HTTPS) provide security services. These protocols help protect data while it is being transmitted across the network by encrypting traffic and preventing unauthorized access. TLS secures communication by ensuring confidentiality and data integrity, while HTTPS uses TLS to protect web traffic between a web browser and a web server.

Together, these protocols help prevent sensitive information from being intercepted or tampered with by using encryption keys to secure communication between parties. This level of protection is especially important for safeguarding FIC Bank's system and sensitive data. According to Cloudflare, HTTPS relies on a public and private key system, where the private key is controlled by the website owner and the public key is available to users who need to securely communicate with the server. This process helps maintain secure communication within a financial organization's network.

Exercise 4: Validate remote access control configuration to be aligned with security policies

The following assessment is based on the Remote Access Policy developed by for FIC Bank in Unit 1:

1.0 Purpose

The purpose of this policy is to explain the basic networking concepts that support secure remote access for FIC Bank. As a global financial institution planning to expand operations to Kiev, Ukraine, FIC Bank must allow employees and administrators to connect remotely while continuously protecting sensitive financial data and internal systems.

Remote access introduces significant security risks, as users connect from external networks that are not controlled by FIC Bank. Understanding how networking protocols operate and how remote access connections are structured helps reduce the risk of unauthorized access, data interception, and system compromise (Buckman, 2025). Overall, this purpose explains why the use of secure networking protocols and controlled remote access is important for maintaining FIC Bank's security posture.

This remote access policy follows the structure outlined in the NIST Remote Access Policy Template which clearly defined access conditions, approved connection methods, and enforcement mechanisms to reduce organizational risk associated with remote connectivity. In addition, this policy reflects the expected remote access topology for a financial institution, where remote desktop and administrative connections such as RDP and SSH are only permitted after users are authenticated and placed into an encrypted tunnel through the bank's VPN gateway.

2.0 Policy

FIC Bank permits remote access to its network only for authorized employees, administrators, and approved third parties who have a legitimate business need. All remote access must follow established security requirements to protect FIC Bank's systems and sensitive financial data.

Remote access must use approved, updated, and secured methods. Employees are required to follow strict rules and guidelines before remote access connections are permitted. According to Cisco, these guidelines include the use of secure encryption through a remote access VPN, strong authentication with multi-factor authentication (MFA), regularly updated security protocols, endpoint security software, and secure service edge (SSE) technologies to protect access to network resources.

FIC Bank's system must be granted based on defined user roles and job responsibilities to gain access. Role-based access controls help ensure that users can access only the systems and data necessary to perform their job functions, which limits potential damage in the event of a security incident. Remote access activity must be continuously monitored to detect unusual or unauthorized behavior and support timely security response.

Users accessing FIC Bank's network remotely must comply with all security policies and are responsible for protecting their login credentials. Sharing credentials, using unapproved remote access tools, or attempting to bypass security controls is strictly prohibited.

This policy ensures that remote access is provided in a consistent and secure manner while reducing the risk of unauthorized access, data exposure, and cyberattacks.

3.0 Connection Procedures

FIC Bank's network requires all remote access users to follow an approved and secure connection process that uses a controlled network layout. Remote users are not permitted to connect directly to internal systems from the public internet. This requirement helps protect FIC Bank's internal network and sensitive financial data from unauthorized access.

Before reaching internal resources, all remote connections must pass through a centralized remote access topology where secure entry points are used to enforce security controls. This topology follows a structured path of remote user, internet, VPN, internal network, ensuring that security checks occur before any access to internal systems is granted. By using a centralized and controlled connection process, FIC Bank is able to verify user identity, apply encryption, and monitor remote access activity in a consistent manner (NIST SP 800-46)

The approved remote access connection process at FIC Bank is as follows:

1. A remote user initiates a connection from an external network, such as a home network or remote office location.
2. The user connects to FIC Bank's approved remote access VPN, which serves as the secure entry point into the network.
3. The VPN verifies the user's identity using approved credentials and multi-factor authentication (MFA).
4. Once authentication is successful, an encrypted connection is established to protect data as it travels between the remote user and FIC Bank's network.
5. After passing through the secure entry point and internal security controls, the user is granted access only to the internal systems and resources they are authorized to use, such as remote desktop or secure administrative services SSH or approved administrative tools.

All remote access sessions must be disconnected when no longer in use. Following these connection procedures ensures that remote access to FIC Bank's network is consistent, secure, and protected against unauthorized access and cyber threats.

4.0 Compliance

All employees, administrators, contractors, and approved third parties who are granted remote access to FIC Bank's network must comply with the Remote Access Policy. Compliance with this policy is required to protect FIC Bank's systems, sensitive financial data, and internal network resources.

Compliance assessments will be conducted to ensure alignment with industry best practices, including NIST SP 800-46 and applicable financial industry security standards. Compliance is

enforced through centralized authentication controls, access logging, and periodic security reviews to detect violations and ensure continued adherence.

Any user or device found to be in violation of this policy may have remote access privileges suspended or revoked until the issue has been resolved. Failure to comply with this policy may increase the risk of unauthorized access, data exposure, or cyberattacks. Ensuring compliance helps FIC Bank maintain a secure network environment and meet regulatory and security obligations.

5.0 Exceptions to Policy

Exceptions to this Remote Access Policy may be granted only under limited and approved circumstances. Any exception must be based on a documented business need and approved by FIC Bank's cybersecurity or IT management team before access is granted.

All approved exceptions must include appropriate security controls to minimize risk and must be reviewed on a regular basis. Exceptions are temporary and may be revoked at any time if they are found to increase security risk or if the business need no longer exists.

Unauthorized exceptions or failure to follow approved exception requirements are not permitted and may result in disciplinary action or removal of remote access privileges. This approach ensures that any deviations from the policy are carefully controlled while maintaining the overall security of FIC Bank's network and sensitive data.

6.0 Other Applicable Policies

The Remote Access Policy is supported by other FIC Bank security policies that help protect information systems, network resources, and sensitive financial data. These policies work together to ensure that remote access is used safely and responsibly when users connect to FIC Bank's network from external locations.

Applicable policies include FIC Bank's Acceptable Use Policy, Password and Authentication Policy, Data Classification and Encryption Policy, and Incident Response Policy. The Acceptable Use Policy (AUP) defines appropriate behavior and system usage expectations for all users. One benefit of implementing an AUP is that it demonstrates "due diligence" and helps protect the organization from reputational harm (Cowbell, 2023).

The Password and Authentication Policy establishes requirements for protecting login credentials, including password standards and the use of multi-factor authentication. Using multiple authentication policies allows organizations to test security controls with a smaller group of users before implementing them across the entire organization, reducing the risk of operational disruption (Tang, 2021).

The Data Classification and Encryption Policy explains how sensitive financial data must be protected during storage and transmission. The Incident Response Policy outlines the actions that must be taken if a security incident occurs during remote access activity.

Remote access allows users to interact with internal systems and perform work as if they were physically connected to the organization's network. While this capability supports business operations and flexibility, it also introduces additional security risks when access originates from outside the network perimeter. Supporting policies help define user responsibilities, enforce access limitations based on job roles, and reduce the risk of unauthorized access or data exposure associated with remote connectivity (Fortinet, 2024)

As a financial institution, FIC Bank must ensure that its remote access practices align with industry security expectations and regulatory requirements for protecting customer financial information. Supporting policies reinforce the use of strong authentication, encryption, and access controls to maintain data confidentiality and system integrity. Together, these policies help FIC Bank maintain a consistent and secure approach to remote access across the organization (Splashtop, 2024). Together, these policies reinforce a defense-in-depth appropriate for financial institution handling regulated customer data.

Failure to comply with any applicable policy may result in disciplinary action or the suspension or removal of remote access privileges.

7.0 Applicability

The Remote Access Policy applies to all FIC Bank employees, administrators, contractors, consultants, vendors, and approved third parties who are granted remote access to FIC Bank's network, systems, or data.

The policy applies to all approved remote access methods, including remote access VPN connections, remote desktop services, and secure administrative access tools. It applies to all devices used to access FIC Bank's network remotely, whether the devices are organization-owned or approved personal devices.

This policy applies to remote access connections originating from any external location, including home networks, remote offices, and travel locations. All users within the scope of the policy are required to comply with its requirements to ensure that remote access is used in a secure and controlled manner.

- Secure Configuration means keeping remote access systems securely configured and up to date helps reduce known vulnerabilities. NIST recommends regularly applying security updates to remote access infrastructure. For FIC Bank, this reduces the risk of attackers exploiting outdated software to gain access to financial systems.

By using VPN, MFA, access controls, endpoint security, firewalls, monitoring, and secure system configurations, FIC Bank can reduce the risks associated with remote access. These network defense components work together to protect sensitive financial data and support secure remote access in line with NIST SP 800-46r2 guidance.

Assessment of Policy Implementation and Alignment With Best Practices

FIC Bank's Remote Access Policy was implemented to secure remote access in alignment with established security best practices. The policy requires all remote connections to pass through an approved VPN, which establishes an encrypted tunnel before users are granted access to internal

systems. FIC Bank handles sensitive financial data, which requires additional protection to prevent information from being exposed over public or untrusted networks.

In addition, all users are required to use multi-factor authentication (MFA) to strengthen the identity verification process. Requiring more than just a password helps reduce the risk of unauthorized access if credentials are compromised. These security controls align with NIST SP 800-46 recommendations, which emphasize encryption, secure configuration and patching of remote access servers, strong authentication, and centralized access points for secure remote connectivity (Centraleyes, n.d.).

There are more best practices that the policy implemented, including role-based access control and continuous monitoring. By incorporating these practices, the policy helps reduce the risk of attack in remote access environments. FIC Bank has limited and structured access based on job responsibilities. By limiting the access, users will only be allowed to access the systems and data necessary for their roles. Continuous monitoring and logging of remote sessions support early detection of suspicious or unauthorized activity, allowing for faster incident response. Requiring secure configuration management and regular updates further reduces the likelihood of attackers exploiting known vulnerabilities. With the implementation of the policy, these controls provide a set of guidelines and strategies that enhance the overall protection of the FIC Bank's systems and data from remote access threats (Centraleyes, n.d.).

Alignment with Financial Sector Remote Access Control Requirements

FIC Bank's Remote Access Policy aligns with security expectations within the financial sector by focusing on protecting sensitive financial data and maintaining strict control over who can access internal systems. As a financial institution, FIC Bank must implement strong endpoint security to act as an additional layer of protection against attackers attempting to exploit remote access services. Endpoint security helps ensure that employee devices meet security requirements before connecting to the network. According to OptionOne, implementing endpoint security allows organizations to deploy antivirus and anti-malware software across employee devices, which reduces the risk of compromised systems accessing internal resources.

Another important access control requirement for the financial sector is secure remote collaboration. As FIC Bank continues to expand internationally, employees will rely on remote access for virtual meetings, cloud storage, and secure messaging. It is important to implement secure video conferencing solutions, encrypted cloud platforms, and protected communication tools to ensure that sensitive financial information is not exposed during remote collaboration. These controls allow employees to work efficiently while maintaining data confidentiality and integrity.

In addition, continuous monitoring and an established incident response plan are critical for financial institutions. Security teams must actively monitor remote access activity to detect potential security breaches and respond quickly when suspicious behavior occurs. By combining endpoint security, secure collaboration tools, and ongoing monitoring, FIC Bank's remote access configuration aligns with financial-sector security expectations and supports compliance with industry best practices. By combining endpoint security, secure collaboration tools, and ongoing

monitoring, FIC Bank's remote access configuration aligns with financial-sector security expectations and supports compliance with industry best practices.

Exercise 5: Have you used a VPN before on a Wi-Fi network? Knowing what you do now about secure remote access controls, are there times you should use a VPN in your life?

I have used a VPN on a Wi-Fi network before, but I did not fully understand its purpose at the time. When I worked at Chinese Laundry as a Merchandising Assistant, my role involved operating and monitoring the wholesale website. I managed inventory, website images, and overall site functionality, and I regularly handled sensitive documents and stored company information across multiple files. Before learning about secure remote access, I assumed the VPN was mainly used to allow employees full access to company systems. I did not fully consider the security and data protection benefits of using a VPN, such as encrypting data in transit and protecting credentials from interception on untrusted networks.

During the COVID-19 pandemic, we transitioned to remote work very quickly, and there was limited time to understand the cybersecurity aspect of remote access. The company provided basic training on topics such as phishing awareness, but I primarily viewed the VPN as a tool that allowed me to work as if I were still in the office. It allowed me to access internal systems and files from my personal laptop, which made my workload more manageable. Now that I understand how VPNs protect data by encrypting network traffic and reducing exposure on public or home Wi-Fi networks, I recognize their importance in a professional setting.

Based on what I have learned, a VPN should be used in situations where sensitive information is being accessed, especially in work environments or small businesses that rely on remote access. For personal use, I believe basic firewall protection is essential, but a VPN becomes more important when accessing sensitive accounts, public Wi-Fi networks, or work-related systems. Overall, VPNs play a critical role in protecting data and maintaining secure remote access, particularly for organizations that handle confidential information.

References:

- What is cryptography? Key concepts for cybersecurity leaders - SecurityScorecard. (2025, July 28). SecurityScorecard. <https://securityscorecard.com/blog/what-is-cryptography-key-concepts-for-cybersecurity-leaders/>
- Longbottom, C. (2018d, August 16). Establish a data classification model for cloud encryption. Search Cloud Computing. <https://www.techtarget.com/searchcloudcomputing/tip/Establish-a-data-classification-model-for-cloud-encryption>
- Greene, N., & Greene, N. (2025, August 19). 16 Most common network protocols. Auvik. <https://www.auvik.com/franklyit/blog/common-network-protocols/>
- Hypertext Transfer Protocol HTTP. (n.d.). F5, Inc. <https://www.f5.com/glossary/hypertext-transfer-protocol-http>
- Dor. (2021, December 12). NIST 800-46. Centraleyes. <https://www.centraleyes.com/nist-800-46/>
- Understanding Firewalls for Home and Small Office Use. (2023, February 23). Cisa. <https://www.cisa.gov/news-events/news/understanding-firewalls-home-and-small-office-use>
- What is Role-Based Access Control (RBAC)? Examples, Benefits, and More. (2025, July 31). Fortra. <https://www.fortra.com/blog/what-role-based-access-control-rbac-examples-benefits-and-more>
- Souppaya, M., Scarfone, K., National Institute of Standards and Technology, & Scarfone Cybersecurity. (2016c). NIST Special Publication 800-46 Revision 2 Guide to Enterprise Telework, Remote Access, and Bring Your Own Device (BYOD) Security [Report]. <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-46r2.pdf>
- Chapter 9: Securing remote access. (2012, June 27). Infosecinstitute. <https://www.infosecinstitute.com/resources/general-security/securing-remote-access/>
- Comparing Different Firewall Types. (n.d.). Cdw. <https://www.cdw.com/content/cdw/en/articles/security/firewall-type-comparison.html>
- Secure Remote Access in 2025: Risks, protocols & best practices. (2025, December 14). Seraphic Security. <https://seraphicsecurity.com/learn/secure-remote-access/secure-remote-access-in-2025-risks-protocols-and-best-practices/>
- Option One. (2023, August 30). *Embracing Remote work: Cybersecurity best practices for financial firms - Option One*. <https://optiononetech.com/insights/embracing-remote-work-cybersecurity-best-practices-financial-firms/>